

Chapter 4

Theory of Low Temperature Gas-Phase Reactions

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4.1. Introduction

In this chapter, we describe the current status of theoretical kinetics for chemical reactions at low temperature, i.e., from ~ 1 to ~ 200 K. The desire to understand the chemistry of interstellar space and of low temperature planetary atmospheres provides the general motivation for studying chemical kinetics at such temperatures. For example, the chemistry of Titan's atmosphere is currently a topic of considerable interest. This motivation led to the development of novel experimental techniques, such as the CRESU (cinetique de reaction en ecoulement supersonique uniforme) method,¹ which allows for the measurement of rate coefficients at temperatures as low as 10 K (see Chapter 2 by Canosa *et al.*). Such measurements provide important tests for theory and have sparked a renewed interest in theoretical analyses for this temperature range.²

Experimental studies of low temperature kinetics are generally limited to reactions that are quite rapid. Furthermore, only such rapid reactions are generally of importance in low temperature chemical environments. Rapid low temperature reactions generally have no barrier (or only one that lies below reactants) along some reaction pathway that leads to the production of exothermic products. For such reactions, it would seem appropriate to focus on the kinetics of the capture process while considering only the long-range contributions to the interaction potential. Indeed, for ionic reactions, this long-range capture rate often provides an adequate description of the low temperature experimental observations. However, for neutral reactions, perhaps somewhat surprisingly, the experimentally observed rate coefficients are seldom in agreement with such long-range capture rate predictions, often being considerably lower than expected.^{3,4}

In some cases, the reduction in the rate coefficient may be related to reflection by a low entropy transition state in the exit channel.⁵ Another general explanation, which is confirmed by detailed *ab initio* transition state theory calculations,⁶⁻⁹ is that the decrease in entropy that occurs with decreasing separation yields an "inner" transition state. This inner transition state can significantly reduce the entrance flux below that predicted by the long-range capture process, even at temperatures as low as a few K. The properties of this inner transition state are determined by a delicate balance between chemical bonding interactions and short-range repulsions, both of which increase in magnitude with decreasing separation. Importantly, the location of this inner transition state is generally distinct (i.e., well separated) from that for the long-range capture process. As a result, the effects of this inner transition state can be significant even when the

long-range (or “outer”) transition state lies at a separation where bonding and repulsion interactions are irrelevant.

Theoretical studies of the long-range capture rate have a long history.^{4,10,11} A combination of classical and quantum statistical theory analyses, as well as classical and quantum dynamical simulations, have yielded a clear qualitative and quantitative understanding of this process. Only the contribution of nearly degenerate electronic states remains somewhat poorly understood. Otherwise, it is relatively straightforward to predict the long-range capture rate to within $\sim 10\%$. The present review begins with a pedagogical summary of the contributions from the different theoretical methods to our understanding of long-range capture rates.

Theoretical efforts at quantitatively predicting the effects of short-range interactions on low temperature rate coefficients have been more limited. This neglect is due in part to the perceived lack of importance of these effects and in part to the difficulty of their study, which must rely heavily on *ab initio* electronic structure calculations. Fortunately, the remarkable advances in *ab initio* quantum chemistry, which are due to methodological developments as well as to the ever increasing computational capabilities, now allow for the quantitative study of the inner transition state region. Indeed, the coupling of these advances in quantum chemistry with continuing developments in transition state and related kinetic theories have transformed theoretical chemical kinetics from a simply interpretive and modeling tool to a truly predictive one. The next section briefly discusses the application of *ab initio* simulations to the prediction of rate coefficients for rapid low temperature reactions.

Subsequently, the inner transition state, and its importance for different classes of reactions, is reviewed. The radical-molecule, radical-radical, and ion-neutral cases are each discussed in some detail. A two transition state model, which accounts for the combined effect of the inner and outer transition states acting in series, is summarized. Both adiabatic and statistical models for the electronic motions are also briefly discussed.

Ab initio based transition state theory studies are used to illustrate the transition from long-range dominated capture to short-range dominated capture for sample reaction classes. The presence, or lack, of two minima in the reactive flux is demonstrated with illustrative plots for each class of reaction. With modest adjustments in saddlepoint energies, the predicted rate coefficients are seen to be in quantitative agreement with CRESU experimental results for a number of reactions. This success is a testament to the high accuracy of current *ab initio* methods as well as the underlying kinetic assumptions. These illustrative applications also highlight the

need for an improved understanding of non-adiabaticities in (i) the electronic states at long range and (ii) the dynamics between the two transition states.

The review closes with a brief summary of the current status of theoretical treatments for both long-range and short-range transition states.

4.2. Capture Rates for Long-Range Potentials

Classical statistical theories provide the simplest procedures for predicting the capture rate. They continue to be widely employed because they are of sufficient accuracy for applied purposes and require significantly less computational resources than dynamical theories. They also provide a useful reference framework for more accurate dynamical theories. Statistical theories are particularly useful in predicting the energy and angular momentum dependence of the reactive flux for use in the calculation of rate constants for multichannel reactions.

Classical trajectory simulations provide accurate predictions of the capture rate as long as (i) an accurate interaction potential is employed, (ii) the temperature is not too low, and (iii) the reaction occurs on only a single electronic state or the electronic dynamics is adiabatic. Such simulations have provided important validation tests for classical statistical theories. Parameterized fits provide useful representations of the trajectory results for a number of specific interaction potentials.

Quantum mechanical approaches are required for low temperatures (e.g., <10 K), where the quantization of rotational and electronic energy levels becomes important. Quantization of translational motion is only of importance at very low temperatures ($\sim 10^{-3}$ K). Fortunately, the vibrational motions of the reactants factor out of the problem, and so the quantization of vibrations is irrelevant. Quantum dynamical studies of some limiting cases provide the most definitive tests for low temperature. In many instances, adiabatic channel approaches, which can be viewed as quantized statistical approaches, provide a sufficiently accurate treatment of the quantum effects. Fortunately, since there are relatively few accessible energy levels at low temperatures, such quantum approaches become readily applicable at precisely the temperature where quantum effects become important.

Taken together, the combination of classical and quantum statistical and dynamical approaches yields a quantitative description of the capture process. The following subsections review the contributions of each of these methods to our understanding of the capture rate.

4.2.1. Statistical Theories for Isotropic Potentials

The phase space theory (PST)^{12,13} assumption of an isotropic interaction between two reacting fragments greatly simplifies the theoretical analysis of their capture process. The capture problem then reduces to the central force problem on an effective potential. This effective potential, $V_{\text{eff}}(R)$, is given by the sum of the isotropic long-range interaction potential, $V_{\text{LR}}(R)$, and the orbital angular momentum, l , dependent centrifugal energies:

$$V_{\text{eff}}(R) = V_{\text{LR}}(R) + \frac{\hbar^2 l(l+1)}{2\mu R^2}, \quad (4.1)$$

where R is the distance between the centers-of-mass of the two reacting moieties, μ is the reduced mass, $m_1 m_2 / (m_1 + m_2)$, and $\hbar$ is Planck's constant, h , divided by 2π . Writing the isotropic interaction potential as $-C_n/R^n$, yields l -dependent centrifugal barriers, $E_l^\dagger$, given by

$$E_l^\dagger = \left(\frac{n}{2} - 1\right) C_n \left[\frac{\hbar^2 l(l+1)}{\mu n C_n} \right]^{n/n-2}. \quad (4.2)$$

A classical treatment of the central-force problem implies that the probability of reaction for each of the reactant states is simply a step function in the translational energy relative to the centrifugal barrier height. The capture rate for a given total energy, E , and total angular momentum, J , is then proportional to the number of states, $N(E, J)$, whose translational energy exceeds $E_l^\dagger$. For the most general case of two non-linear fragments this "phase space theory" number of available states is given by

$$N(E, J) = (2J + 1) \sum_{j_1, k_1, j_2, k_2, l, j, v_i} \Theta(E - E_{j_1 k_1} - E_{j_2 k_2} - E_{v_i} - E_l^\dagger) \\ \times \Delta(J, j, l) \Delta(j, j_1, j_2) \quad (4.3)$$

where $E_{j_i k_i}$ is the rotational energy of fragment i in rotational state $j_i k_i$, E_{v_i} is the vibrational energy of the fragments in vibrational state v_i , Θ is the step function, and Δ is the triangle inequality.

The thermal capture rate may be written as

$$k(T) = \frac{1}{hQ} \int N(E, J) \exp(-\beta E) dE dJ, \quad (4.4)$$

where Q is the canonical partition function for the reactants including relative translation, and $\beta = 1/k_B T$, with T as the temperature, and k_B as Boltzmann's constant. If the vibrations of the fragments are assumed to be decoupled from the reactive motions, then the sums over vibrational states in Eq. (4.3) cancel with the corresponding contribution to the canonical

partition function of the reactants in Eq. (4.4) and so can be removed from further consideration.

The expression in Eq. (4.3) assumes quantized rotational states. Alternatively, the sum over quantum states may be replaced with a classical phase space integral:

$$N(E, J) = \frac{\hbar}{h^\nu} \int d\Omega_{12} d\Omega_1 d\Omega_2 dP_{\Omega_{12}} dP_{\Omega_1} dP_{\Omega_2} \\ \times \Theta[E - V_{LR}(R^\dagger) - T_{12}(R^\dagger) - T_1 - T_2] \delta(\hat{J} - J\hbar), \quad (4.5)$$

where the contribution from the vibrational states of the fragments has been removed. In Eq. (4.5) ν is the number of rotational degrees of freedom, Ω , P_Ω , and T , denote the orientational coordinates, conjugate momenta, and kinetic energy, respectively, for the orbital motion (subscript $_{12}$) and the reacting fragments (subscripts $_1$ and $_2$), and $R^\dagger$ denotes the value of R correlating with the maximum in the effective potential, Eq. (4.1). This classical expression, Eq. (4.5), provides the conceptual basis for variational transition state theory treatments, whereas the quantum expression, Eq. (4.3), provides the conceptual basis for adiabatic channel treatments.

The PST assumptions of isotropic interactions and corresponding rovibrationally adiabatic dynamics are particularly appropriate for the reaction of ionic species with nonpolar molecules, for which the ion induced dipole interaction is the dominant long-range interaction. For a spherical molecule, the ion induced dipole interaction (in CGS units) takes the simple form:

$$V_{LR}^{iid}(R) = -\frac{\alpha q^2}{2R^4}, \quad (4.6)$$

where α is the polarizability of the molecule, and q is the charge on the ion. For this potential, the PST calculated thermal capture rate is equal to the classical Langevin expression¹⁴

$$k(T) = 2\pi q \sqrt{\alpha/\mu}. \quad (4.7)$$

The generally non-spherical nature of a molecule correlates with asymmetries in the polarizability, and thus in the interaction potential. However, the effect of this asymmetry tends to be quite minor^{15,16} and the simple replacement of the polarizability, α , with the mean polarizability $(\alpha_{||} + 2\alpha_{\perp})/3$ is quite effective. The polarizability can generally be estimated quite accurately as a sum of bond polarizabilities.¹⁷ Alternatively, accurate values can be obtained from straightforward electronic structure calculations (e.g., density functional theory or second order Moller Plesset perturbation theory, MP2, calculations with augmented double-zeta basis sets or larger).

The ion induced-dipole potential and the corresponding Langevin rate expression have proven to be of great value in the interpretation of numerous studies of ion-molecule reaction kinetics (see chapters 2 and 3 by Canosa *et al.*, and by Gerlich). For neutral reactions, a $1/R^6$ potential, which can be taken to represent dispersion and/or dipole induced-dipole interactions, has provided a similarly important reference potential. The Gorin model,¹⁸ which is based on PST like assumptions for this potential, provides a simple expression for the capture rate:^{4,19}

$$k = \Gamma(2/3) 2^{11/6} \pi^{1/2} \frac{C_6^{1/3} (k_B T)^{1/6}}{\mu^{1/2}} \quad (4.8)$$

where C_6 is the coefficient in the potential. The ‘modified’ Gorin model, which has been used, especially by Benson and Golden,^{20,21} to treat barrierless radical-radical reactions, incorporates an empirical steric reduction factor to account for the nonreactivity of certain orientations of the fragments.

4.2.2. Statistical Theories for Anisotropic Potentials

Unfortunately, the assumption of an isotropic interaction seldom holds for either ionic or neutral reactions. For example, for an ion-molecule reaction, the presence of a dipole, d , in the neutral fragment implies that the dominant long-range term is the ion-dipole term,

$$V_{LR}^{id}(R) = -\frac{qd \cos \theta}{R^2}, \quad (4.9)$$

which is clearly anisotropic. The dominance of this term at long-range implies that it is the determining factor for the kinetics in the low temperature limit. Furthermore, even in the absence of a dipole, the anisotropic ion-quadrupole interaction, which is essentially always present, has a $1/R^3$ behavior and thus becomes the dominant interaction at low temperature.

Similarly, for neutral reactions, the dipole-dipole interaction with a $1/R^3$ form, or the dipole-quadrupole interaction with a $1/R^4$ form, or even the quadrupole-quadrupole interaction with a $1/R^5$ form, are each of longer range than the dispersion and dipole induced-dipole terms that contribute to the $1/R^6$ potential. Thus, the isotropic assumptions of PST are essentially never valid when considering low temperature kinetics. At higher temperatures chemical bonding interactions and short range repulsions, which are even more highly anisotropic, become important. As a result, at least for neutral reactions, there is essentially no temperature range where PST is generally applicable. For ionic reactions, the relevance of PST and the corresponding Langevin rate prediction at higher temperatures (e.g. near

room temperature) depends on the magnitude of the polarizability relative to the dipole and/or quadrupole moments. For example, if $d^2/(2\alpha k_B T) < 1$ the deviations from the Langevin predictions can be quite modest.²²

These anisotropies lead to a coupling of the fragment rotational and orbital angular momenta. The quantities l , j_1 , j_2 , k_1 , and k_2 are no longer strictly conserved. This coupling is manifested in the addition of an orientational dependence to $V_{LR}(R)$ in Eq. (4.5) or equivalently to the replacement of $E_{j_1 k_1} + E_{j_2 k_2} + E_l$ in Eq. (4.3) with the energy $E_i(R)$ for the fully coupled motion. There have been numerous attempts to generalize PST to account for the effects of this coupling, with these treatments based on either Eq. (4.3) or Eq. (4.5). Here we summarize the work based on Eq. (4.5), with a discussion of adiabatic capture treatments left for Section 4.2.4.

Much of the work at treating the effects of anisotropies in the potential has focused on ion-molecule reactions, and particularly on the incorporation of corrections for the ion-dipole and/or ion-quadrupole interactions. Various “effective potential” approaches have employed an isotropic potential that is obtained from some sort of averaging of the real anisotropic potential. The average dipole orientation approach of Bowers and coworkers²³ is based on a dynamical average of the orientational angle. In the extended Langevin approach²⁴ the capture rate is instead calculated for fixed angle and then averaged over angle. Markovic and Nordholm discussed the relation of the free energy based effective potential approach²⁵ to canonical variational transition state theory and incorporated microcanonical based restrictions on the energies.²⁶ Most recently, Rayez and coworkers have considered a simple orientationally averaged effective potential.²⁷

Variational transition state theory (VTST) provides a rigorous upper bound to the capture rate and also allows for the simple incorporation of energy and angular momentum conservation. Chesnavich and coworkers developed a sophisticated VTST treatment for the ion-dipole + ion-induced dipole case.²² In a series of articles Niedzielski and coworkers generalized and extended this work to the detailed consideration of a number of specific ion-molecule cases.^{28,29} Smith and Troe applied a VTST treatment, based on the assumption of $J \approx l$, to the potential given as the sum of the ion-dipole, ion-quadrupole, and anisotropic ion-induced dipole potentials.³⁰ Hase and coworkers considered the application of traditional reaction path Hamiltonian approaches, which assume harmonic vibrators for all the vibrational modes including the low frequency interfragment bending motions.³¹

A very general approach was considered by Wardlaw and Marcus,³² who proposed the evaluation of the number of states in Eq. (4.5) via Monte Carlo integration, but with the inclusion of the full orientation dependent V_{LR} . With the advances in computational abilities at the time, this allowed

for the straightforward prediction of the capture rate for arbitrary potentials and for arbitrary rotors (atomic, linear, spherical, symmetric, or asymmetric). This “flexible” TST approach was readily applicable to arbitrary long-range potentials for both ion-molecule and radical-radical reactions. Although very general, the flexible TST approach did not readily yield analytic expressions for the rate coefficients, which is one of the great values of many of the earlier VTST studies.

In recent work, we have developed a “long-range” form of transition state theory that is readily applied to arbitrary potential forms, but also immediately yields analytic results for many simple potential forms.⁴ These analytic results have previously been derived in a variety of ways. The long-range TST approach clarifies the results that have been obtained in these prior studies. For example, it demonstrates that the inadequacies of some of the early transition state theory studies were largely due to the neglect of the conservation of energy and/or angular momentum.

This “long-range” transition state theory is based on the simple assumption that the orbital moment of inertia is much greater than each of the fragment moments of inertia. This assumption is valid when focusing on the long-range part of the potential since the orbital moment of inertia is proportional to R^2 . Generally, the quantitative validity of this assumption becomes debatable for separations of about 5 Å or less. Coincidentally, long-range expansions of the potential also become inappropriate at this same separation.

With this assumption of a large orbital moment of inertia, and the vibrational decoupling assumption, the capture rate can be written as

$$k(T) = \left(\frac{8\pi}{\mu(k_B T)^\nu} \right)^{1/2} \frac{1}{\Gamma(\nu/2 - 1/2)} \int_0^\infty dE e^{-E/k_B T} \int_0^\infty dX N_x^\dagger(E, X), \quad (4.10)$$

where

$$X = \frac{J^2 \hbar^2}{2\mu}, \quad (4.11)$$

$$N_x(E, X|R) = \langle [E - V(O, R) - X/R^2]^{(\nu-3)/2} \rangle, \quad (4.12)$$

and

$$N_x^\dagger(E, X) = \min_R N_x(E, X|R). \quad (4.13)$$

In these expressions, ν is the dimensionality of the system given by $\nu_1 + \nu_2 + 3$, where ν_i is the number of rotational degrees of freedom for reactant i . Also, the average is taken with respect to the orientational space

$O = (\Omega_{12}, \Omega_1, \Omega_2)$ of the rotors, and its argument is set to zero when the expression is negative.

The above result for the capture rate is based on the proper conservation of energy and total angular momentum. Simpler, more transparent, expressions were also derived for the less accurate microcanonical and canonical cases. In particular, for the microcanonical case the rate is given by

$$k(T) = \left(\frac{8\pi}{\mu (k_B T)^\nu} \right)^{1/2} \frac{1}{\Gamma(\nu/2 + 1/2)} \int_0^\infty dE e^{-E/k_B T} N_E^\dagger(E), \quad (4.14)$$

where

$$N_E(E|R) = R^2 \langle [E - V(O, R) - X/R^2]^{\nu-1/2} \rangle, \quad (4.15)$$

and

$$N_E^\dagger(E) = \min_R N_E(E|R). \quad (4.16)$$

For the canonical case, the rate is simply

$$k(T) = \left(\frac{8\pi k_B T}{\mu} \right)^{1/2} N_T^\dagger, \quad (4.17)$$

where

$$N_T(R) = R^2 \langle e^{-V(O, R)/k_B T} \rangle, \quad (4.18)$$

and

$$N_T^\dagger = \min_R N_T(R). \quad (4.19)$$

Typically, neglecting either the conservation of energy or angular momentum yields an overestimate in the capture rate coefficient of about 5–10%. Notably, however, the conservation of E and J can be much more significant when considering multiple transition states.

In the low temperature limit only the longest ranged term in the expansion is relevant. In this case, analytic expressions for the rate coefficient are readily derived. In particular, for arbitrary single term long-range potentials,

$$V_{LR} = \frac{V_0 f(O)}{R^n} \quad (4.20)$$

the long-range transition state theory expression reduces to

$$k(T) = C_1 \mu^{-1/2} V_0^{2/n} (k_B T)^{1/2-2/n}, \quad (4.21)$$

where C_1 is a constant that is obtained from a well specified orientational average of the long-range potential (*cf.* Eq. 4.12), and V_0 is a dimensional constant characterizing the strength of the potential. The function $f(O)$ describes the orientational dependence of the potential. In general, the constant C_1 depends on the dimensionality of the system, the rotational constants of the fragments, and whether or not energy and angular momentum conservation are accounted for.

The analytic results obtained for a number of standard cases are summarized in Table 4.1.

Unfortunately, these results for the single term long-range expansions of the interaction potential are of very limited practical applicability, particularly for neutral reactions. At temperatures below ~ 10 K the classical treatment of the rotational motions becomes inaccurate. Meanwhile, for neutral reactions at temperatures above ~ 10 K, the long-range transition states tend to be at separations where multiple terms in the long-range expansion are of similar magnitude.⁴ For ionic reactions, the lowest order term in the expansion is often dominant until about 100 K, providing a window of applicability from ~ 10 to ~ 100 K. When multiple terms in the potential are of similar magnitude, the long-range capture rate can readily be evaluated from Eqs. (4.10)–(4.12), but now employing the sum of all the relevant contributions to the potential. Nevertheless, the interplay between the different contributions complicates the dependence on temperature and interaction potential.

The maximum attractive contributions from the ion-dipole, ion induced-dipole, and ion-quadrupole terms to the $\text{H}_3^+ + \text{HCl}$ interaction potential

Table 4.1. Components of the analytic expression for the long-range transition state theory capture rate for various asymptotic expansions of the potential.

Potential	Order n	V_0^a	C_1^b
Ion-Dipole	2	$q_1 d_2$	2.5–2.7
Ion-Quadrupole	3	$q_1 Q_2$	5.3–5.4
Ion-Induced-Dipole	4	$\alpha \ 2ql^2/2$	8.9
Dipole-Dipole	3	$d_1 d_2$	4.8–5.0
Dipole-Quadrupole	4	$d_1 Q_2$	4.4–4.5
Dipole-Induced-Dipole	6	$d_1^2 \alpha_2$	8.7
Dispersion	6	C_6	8.55

^a The terms q_i , d_i , Q_i , α_i , and C_6 represent the molecular charge, dipole moment, quadrupole moment, polarizability for fragment i and the dispersion coefficient, where the latter is often taken to be $3\alpha_1\alpha_2 E_1 E_2 / (2(E_1 + E_2))$, where E_i is the ionization energy of fragment i .

^b The coefficient C_1 is generally within the specified range of values, with the actual value depending on the parameters for the specific system.

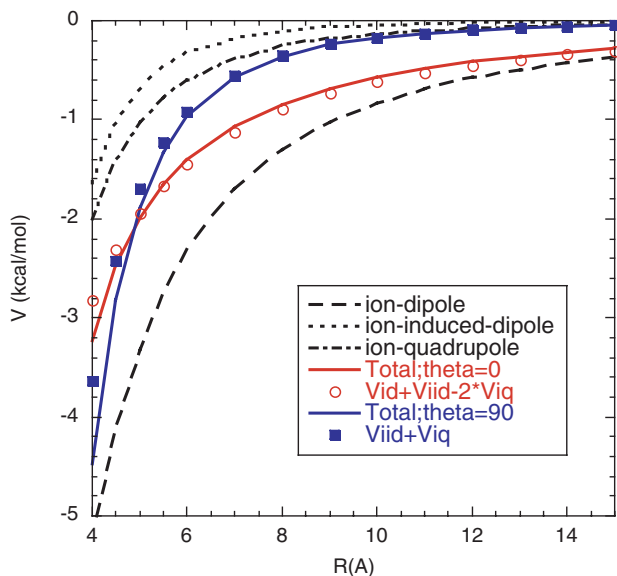


Fig. 4.1. Plot of the contributions to the long-range potential for $\text{H}_3^+ + \text{HCl}$.

are illustrated in Fig. 4.1. The ion-dipole term clearly dominates, but the other two terms approach a similar magnitude as the separation distances R decreases toward 4 Å. Notably, a number of the earlier statistical theory studies of ion-molecule reactions recognized the importance of multiple terms in the long-range potential. Although these studies were generally unable to directly obtain analytic expressions, they did often provide empirical fits to their numerical simulations.

The total potential obtained from MP2/aug-cc-pvtz calculations is illustrated with symbols for dipolar angles of 0 and 90°. The corresponding long-range sums, represented with solid lines, accurately reproduce the *ab initio* potential for separations of ~ 5 Å and larger. The increasing deviations from the long-range representation at separations below ~ 5 Å are expected to be fairly representative for both neutral and ionic reactions.

Explicit prescriptions for the radial dependence of the long-range transition state location were provided in Ref. 4 for a number of representative cases. For ionic reactions, the transition state tends to approach the 5 Å region at about 300–500 K. For neutral reactions, the failures of the long-range estimates are more dramatic and often arise at much lower temperatures due to the much weaker attractiveness of the potential. A more detailed discussion of the failures for neutral reactions, which can extend down to a few K, will be provided in Section 4.4.

4.2.3. Trajectory Simulations

Trajectory simulations of the long-range capture rate were pioneered by Su and coworkers in the 1980's.^{22,33} The reacting fragments were considered as rigid bodies with the analysis considering only the rotational and translational dynamics. These authors focused on ion-molecule reactions, considering the effect of various ion polar interactions on the ion-induced dipole rate coefficients. These trajectory simulations have provided important data for the testing of other more approximate theories. Su and coworkers have also made efforts to parameterize their results³⁴ in order to facilitate their use by other workers. More recently, Troe and coworkers have expanded on this work, avoiding some numerical problems, considering a larger range of parameters, and considering neutral in addition to ionic reactions.³⁵⁻³⁹ They have also provided a parameterization for a large number of cases.¹⁹

One conclusion from these studies is that trajectory simulations are now quite straightforward and can be routinely performed for any analytic potential energy surface. However, there is still interest in statistical theories for a variety of reasons. In particular, when studying multi-channel (or multi-transition state) reactions it is important to have the capture flux at a wide range of E and J . The evaluation of this capture flux from trajectory simulations is still somewhat time-consuming. More importantly, accurate evaluations of the capture rate often require *ab initio* quantum chemical evaluations of the potential energy surface. These *ab initio* evaluations can be quite intensive computationally. The direct coupling of such *ab initio* evaluations with statistical theories requires many fewer potential evaluations than does the corresponding coupling with trajectory simulations. For these reasons it is still important to have some understanding of the quantitative accuracy of statistical predictions for the capture rate. Our own trajectory simulations indicate that the long-range transition state theory predictions for the capture rate are generally accurate to 10% or better.⁴

4.2.4. Adiabatic Channel Calculations and Quantum Dynamics

Adiabatic channel (or equivalently adiabatic capture) calculations provide a generalization of the quantized representation of phase space theory, Eq. (4.3), to account for the effect of anisotropies in the potential.⁴⁰ Such calculations focus on the evaluation of the quantized energy levels, $E_i^J(R)$, for the coupled fragment and orbital rotational motions at a given center-of-mass separation, R , and a given total angular momentum J . The sums over the independent fragment and orbital energy levels are replaced with a sum over the energy levels, i , of the coupled system. For a given energy

E , the reactive flux is given by the number of channels whose adiabatic maxima, $E_i^{J,\max}$ lie below that energy,

$$N(E, J) = (2J + 1) \sum_i \Theta(E - E_i^{J,\max}), \quad (4.22)$$

Notably, adiabatic channel models and variational transition state theory yield identical rate estimates as long as they employ the same assumptions for the energy levels.⁴¹ However, the implementations of the two theories generally employ different assumptions (e.g., classical versus quantum) and so yield modestly different rate estimates. Furthermore, the adiabatic models sometimes implement additional constraints such as an assumed conservation of the orbital angular momentum.

The explicit consideration of the quantized nature of the rotational motions makes adiabatic channel treatments particularly appropriate for examining the kinetics at low temperature. Furthermore, adiabatic channel treatments are most effectively implemented when there are not too many states, i.e., at low temperatures. Classical treatments of the rotational states, as in long-range TST, are accurate when $k_B T/B \gg 1$, where B is the largest rotational constant of the reactants. For typical molecular dimensions, classical partition functions become inaccurate somewhere in the 0.1 to 50 K range.

The accurate evaluation of the quantized rotational energy levels can be a computationally intensive task. For this reason, numerous authors have introduced various approximate schemes. Clary, who pioneered the implementation of adiabatic channel (AC) models for the long-range capture rate, introduced three primary approaches, the ACIOSA, ACPCSA, and ACCSA methods.^{10,42–45} The ACCSA method, which is the most accurate, employs the centrifugal sudden approximation (CSA). This approximation ignores the coupling between different values of the z -projection of the body — fixed total angular momentum. Comparison with close coupled scattering calculations demonstrated the accuracy of the adiabatic capture and centrifugal sudden approximations for the $\text{He}^+ + \text{HCl}$ reaction.⁴⁶ The ACIOSA, which is the least accurate approach, is based on the infinite order sudden approximation (IOSA). This approximation includes the CSA and also ignores the coupling between different angular momentum states of the fragments. With the ACIOSA method, the coupling of the angular momenta reduces to an orientational average, which greatly simplifies the calculations. However, the resulting predictions are often in significant error. The ACPCSA, where P denotes partial, employs the CSA for the fragment with the larger rotational constant, and the IOSA for the fragment with the smaller rotational constant.

Initial applications of the ACCSA method to a number of ionic reactions yielded generally good agreement with experimental observations, but there

were modest discrepancies for a few cases.^{47–49} Subsequent application to a series of first and second row atomic cation reactions found generally good agreement for the first row cations.⁵⁰ For the second row cations discrepancies of a factor of two or three were typical. These and other discrepancies could often be explained at least qualitatively by the presence of high exit channel barriers. Explicit *ab initio* calculations of the potential for a few other reactions indicated modest deviations due to the contribution from additional terms beyond the ion-dipole and ion-induced dipole terms.⁵¹ Unfortunately, these *ab initio* calculations were performed at the SCF level, which does not treat dispersion effects.

Open shell species lead to considerable complications in the analysis related to the presence of multiple electronic states and the effect of rovibronic couplings due to spin-orbit interactions. The separation into ortho and para rotational manifolds have similar effects. Graff and Wagner provided a pioneering study of such effects for the prototypical $O(^3P)+OH(^2\Pi)$ reaction.⁵² Clary and coworkers provided a series of analyses for the reactions of both neutral and ionic $^2\Pi$ radicals.^{49,51,53} These studies focused on the low temperature limit where rovibronic coupling effects become increasingly important. In this limit, only the ground spin-orbit state contributes, and the analysis focuses on the accurate evaluation of the rovibronically coupled energy levels via perturbation theory. This work has been extended by Stoecklin and coworkers to include the reaction of open shell atoms (2P or 3P).⁵⁴ These rotationally adiabatic perturbation theory studies yield useful analytic representations of the zero temperature limit classical adiabatic capture rate for a variety of long-range potentials.

At higher temperatures there can be some contribution from low lying excited electronic states (e.g., a second spin-orbit state). In this case, the multiple surface effects are generally treated via the consideration of the number of surfaces on which the potential is attractive. The single surface rate coefficient is then multiplied by the ratio f_e given by the electronic partition function for the reactive surfaces divided by that for the reactants.⁵⁵ This result corresponds to an adiabatic approximation for the electronic dynamics.

Important further work by Troe and Nikitin and coworkers^{19,30,35–39,56,57} considered the calculation of the capture rate from the perspective of the statistical adiabatic channel model (SACM). Ramillon and McCarroll⁵⁸ demonstrated that the adiabatic capture method of Clary and the SACM method of Troe are identical in concept. However, there are still some minor differences in the approaches used to evaluate the rotational energies. Direct comparison for a number of ion-dipole capture rates found good agreement down to about 50 K, but there were increasing discrepancies at lower temperatures. These discrepancies are apparently a

sign of numerical problems. Related methods include the perturbed rotational state approximation of Sakimoto⁵⁹ and the adiabatic invariance method of Bates and coworkers.⁶⁰

These SACM studies of Nikitin and Troe and coworkers derived general expressions for a wide variety of specific cases and also explored various limitations. Troe's initial focus was on ion-molecule reactions.^{30,35,36,56} Other work focused on linear-linear neutral reactions, initially considering only a dipole-dipole potential.^{37,57} Subsequently, the effects of valence interactions were also considered from a general perspective,^{38,39} and via a detailed study of the OH + OH recombination.⁶¹ These studies each contain useful comparisons with trajectory simulations, with phase space theory calculations, and with zero temperature limiting expressions.

The classical SACM and ACCSA methods fail at very low temperature and sometimes also at intermediate temperatures. In the low temperature limit, both the adiabatic assumption and the classical treatment of the relative translational motion fail. Vogt and Wannier⁶² have shown that the zero energy limit of the quantum cross section for an ion-induced dipole potential is twice that predicted by the classical Langevin model. The classical SACM and ACCSA methods both predict the Langevin result. An empirical study by Hessler⁶³ of the CN + O₂ reaction interpolated from classical high energy cross sections to the Bethe and Wigner threshold behaviors. These empirical calculations suggest that quantum translational effects become important at about 7 K. However, subsequent detailed analyses by Nikitin and Troe and coworkers indicate that the classical SACM model is actually accurate down to $\sim 10^{-3}$ K.^{64,65} For even lower temperatures they have developed an axially non-adiabatic channel approximation and have also performed explicit quantum coupled channel calculations.

The higher temperature failures arise solely from non-adiabatic transitions, with the global Massey parameter, ξ , being a key indicator.¹⁹ This parameter is defined as the ratio of the average collision time to the average rotation time. When $\xi \gg 1$, the dynamics is globally adiabatic. Nikitin and Troe and coworkers have estimated the Massey parameter and non-adiabatic effects for a range of potentials and other parameters. This analysis shows that the dynamics, sometimes, but not always, become non-adiabatic at some temperature that is almost always high relative to the rotational temperatures of the reactants. Then, at some even higher temperature, the dynamics returns to being globally adiabatic. For this intermediate temperature regime, where the dynamics is non-adiabatic, Troe and coworkers have advocated the use of trajectory simulations. In some cases, the dynamics remains adiabatic throughout the regime where classical rotational descriptions are appropriate and the classical SACM models

are applicable. Alternatively, the use of a post-adiabatic representation yields a description in terms of uncoupled dynamic states.⁶⁶

Notably, their discussions about non-adiabaticities have been for a reaction coordinate assumed to be the separation between the centers-of-mass of the two reactants. The use of a more general reaction coordinate, as in some VTST methods, may account for some of the non-adiabaticities. Indeed, reaction coordinate optimizations in the variable reaction coordinate (VRC)-TST approach⁶⁷ have often found a reduction by a factor of two or more relative to that obtained for the separation between the centers-of-mass. However, these reductions are generally for short range separations where the long-range potential expansions are inappropriate.

The adiabatic channel model directly yields estimates for the rate coefficient $k_j(T)$ for the capture rate as a function of the rotational quantum state j of the reactants.⁴² For a given j , such rate coefficients generally increase with temperature. Meanwhile, at a given temperature, the $k_j(T)$ generally decrease with increasing j and the population of high j states increases with increasing temperatures. Due to these two competing factors, the rate coefficient can either rise or decline with increasing temperature depending on their relative importance. Interestingly, a recent experiment of Olkhov and Smith provides detailed information on the j dependence of the capture rate for the reaction of CN with acetylene.⁶⁸

4.2.5. Illustrative Comparisons

Some long-range TST predictions for the $\text{H}_3^+ + \text{HCl}$ capture rate are illustrated in Fig. 4.2. The capture rate obtained by considering solely the ion-dipole potential is in good agreement with that for the total potential, consisting of the sum of the ion-dipole, ion induced-dipole and ion-quadrupole potentials, for temperatures up to about 100 K. At higher temperatures, the other terms in the potential yield an increasing contribution to the long-range capture rate. By 1000 K, the capture rate for the total potential is about twice as large as that for the ion-dipole potential. However, in reality, by 1000 K the long-range capture rate theory is no longer applicable, because the transition state is at too short a separation. Many more potential terms need to be included, and a long-range expansion is no longer effective.

Also shown in Fig. 4.2 are the predictions from ACCSA⁴⁸ and SACM calculations,⁶⁹ and the experimental results from Adams and coworkers.⁴⁸ Above 20 K, the long-range TST and ACCSA results are within 10% of the SACM results, which in turn are in excellent agreement with the experimental results. At lower temperatures the long-range TST results overpredict the SACM results, which are expected to be accurate. This

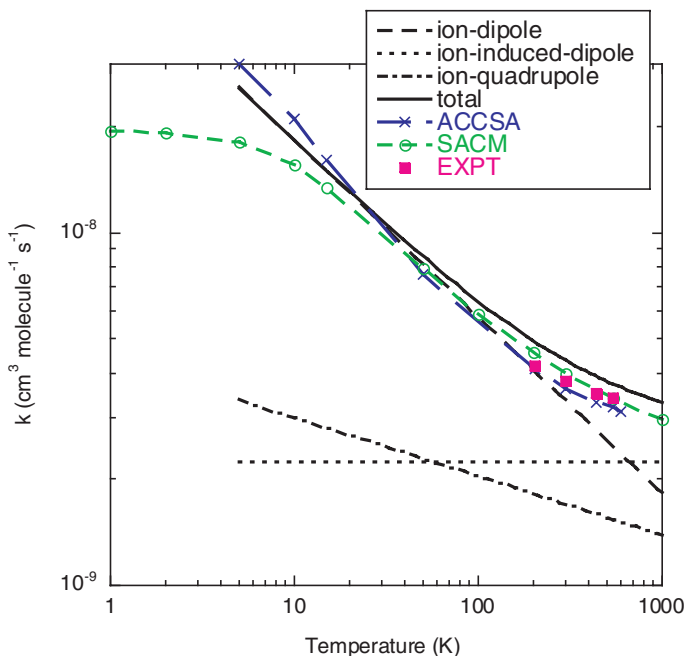


Fig. 4.2. Plot of the temperature dependence of the $\text{H}_3^+ + \text{HCl}$ capture rate for different terms in the potential and for different theoretical methods and for experimental data from Ref. 48.

deviation is likely due to inadequacies in the classical treatment of the HCl and H_3^+ rotational motions, which have rotational constants corresponding to 15 and 63 K, respectively. The low temperature deviations of the ACCSA results from the SACM results arise from numerical inaccuracies in their earliest implementation.

In Fig. 4.3, the long-range TST and SACM⁶⁹ results for the capture rate in $\text{N}^+ + \text{H}_2\text{O}$ are compared with experimental^{70,71} results that extend down to 27 K. Both theoretical predictions are again in good agreement with experiment, with a maximum discrepancy of only 20%. The deviation between the two theoretical predictions at low temperature is again indicative of important quantum effects. The rotational constants for H_2O are equivalent to temperatures of 14, 21, and 39 K.

The quantum effects are complicated by the nearly degenerate $^3\text{P}_0$, $^3\text{P}_1$, and $^3\text{P}_2$, spin-orbit states of N^+ . It is not clear how many electronic states are reactive. The red curve assumes that there are two reactive triplet states, and that the electronic motion is adiabatic. This assumption seems reasonable, given the two unpaired electrons in N^+ . If, instead, only one

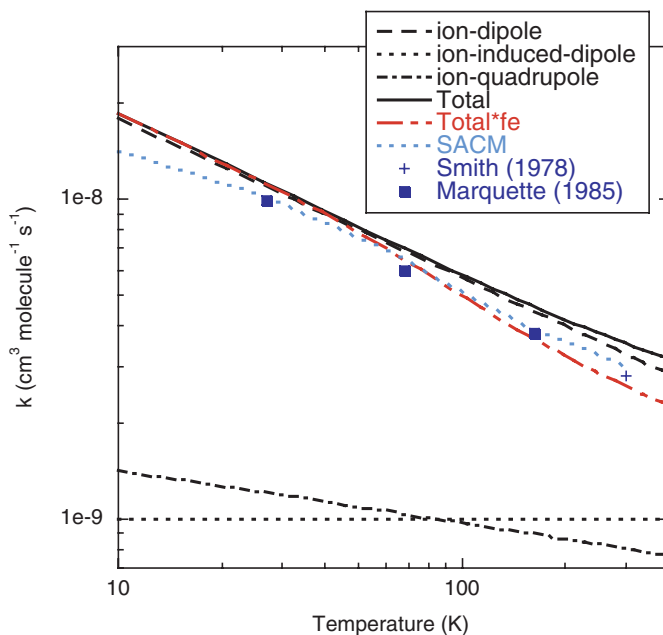


Fig. 4.3. Plot of the temperature dependence of the $\text{N}^+ + \text{H}_2\text{O}$ capture rate for different terms in the potential and for different theoretical methods and for experimental data from Refs. 70 and 71.

triplet state was reactive, the predicted rate would be another factor of two lower at 300 K, well below the experimental result. Such ambiguities in the long-range predictions are unfortunately quite commonplace. The proper treatment of the electronic degeneracy factor f_e , requires detailed *ab initio* simulations of the minimum energy paths, and also hinges on the adiabatic or statistical nature of the electronic dynamics.

CRESU experiments indicate that a number of radicals react with unsaturated hydrocarbons at close to the capture rate, especially at low temperature.⁷² In Fig. 4.4, long-range TST predictions for the $\text{CH} + \text{C}_2\text{H}_2$ capture rate are illustrated together with the experimental data from Refs. 73–75. The prediction for the full potential, given by the sum of the dipole-quadrupole, dispersion, and dipole induced-dipole potentials, already deviates from the dominant dispersion term at 10 K. By 200 K, the dipole-quadrupole term has become the dominant factor. The different contributions to the rate coefficient over the 0 to 20 K region for the $\text{CN} + \text{O}_2$ and $\text{CH(A)} + \text{CO}$ reactions were studied with ACCSA calculations by Tan *et al.*⁷⁶

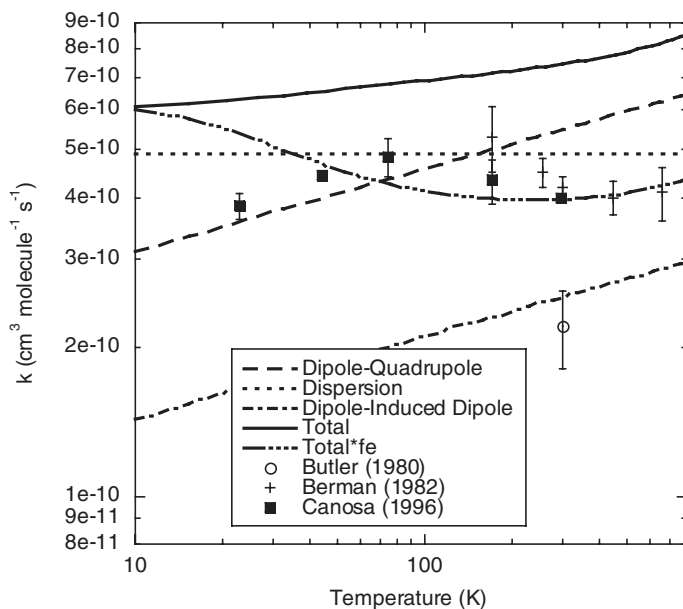


Fig. 4.4. Plot of the temperature dependence of the capture rate for different terms in the potential in comparison with experimental data^{73–75} for $\text{CH} + \text{C}_2\text{H}_2$.

The CH radical has only a 28 cm^{-1} spin-orbit coupling leading to the $^2\Pi_{1/2}$ and $^2\Pi_{3/2}$ states. Sample CASPT2/aug-cc-pvtz//CASPT2/aug-cc-pvdz calculations indicate that only one of the two doublet states has a barrierless reaction path. Thus, the long-range TST predictions should be multiplied by the electronic degeneracy factor f_e given by $2/(2 + 2 \exp(-40/T(\text{K})))$. The dash-dot-dot-dot line corrects the long-range TST prediction for the capture rate by this factor, and is in remarkably good agreement with experiment. The deviations for the lowest temperature, 23 K, are likely indicative of quantum effects given that the rotational constant of CH corresponds to 21 K. Notably, the quantum adiabatic capture rate for the dipole-quadrupole potential, which is the dominant term at low-temperature, is predicted to have a $T^{1/4}$ dependence as $T \rightarrow 0$.⁵³

Similar agreement between theory and experiment has been found for ACCSA calculations of the capture rate for a number of atom diatom reactions. In particular, the predictions for the $\text{B} + \text{O}_2$,⁷⁷ $\text{Al} + \text{O}_2$,⁷⁸ $\text{Si} + \text{O}_2$ and NO ⁷⁹ reactions accurately reproduce the temperature behavior and the numerical values are generally within about 30% of the experimental data. The incorporation of spin-orbit effects on the rovibronic energy levels is a key component of these studies. For the $\text{C} + \text{NO}$ reaction, trajectory

simulations⁸⁰ are in good agreement with experiment,⁸¹ whereas ACCSA predictions⁸² appear to be too low by about a factor of 3. Some of the improvement in the trajectory result arises from the replacement of the form for f_e employed in the ACCSA treatment with an adiabatic form.

Other classes of neutral reactions that appear to proceed at close to the capture rate over the full low temperature range (10 to 200 K) include the reactions of atomic B(³P),^{83,84} C(³P),^{85,86} and Si(³P)⁸⁷ atoms with C₂H₂ and C₂H₄ and the reactions of CH,^{73,88,89} and CN^{90,91} with unsaturated hydrocarbons ranging from acetylene and ethylene to anthracene. The reactions of C₂H with unsaturated hydrocarbons⁹²⁻⁹⁵ appear to proceed at close to but not quite the capture rate.⁴

4.3. *Ab Initio* Simulations

The quantitative exploration of the effect of short-range interactions on the low temperature kinetics requires detailed *ab initio* simulations. We have recently provided an extensive review of *ab initio* methods for studying reactive potential energy surfaces.⁹⁶ Here, for completeness, we provide a brief review of some of the key aspects relevant to the treatment of rapid low temperature reactions.

4.3.1. *Radical-Molecule Systems*

4.3.1.1. *Inner Transition State*

For radical-molecule reactions the number of states at the inner transition state is largely determined by the energy and rovibrational properties of the saddle point. Simple rigid-rotor harmonic oscillator evaluations at the saddlepoint appear to yield an adequate description of the number of states at the inner transition state. Explicit consideration of variational effects and barrier tunneling for some sample reactions found them to be of at most modest importance as long as the reaction is rapid.^{7,8} Fully coupled phase space integral based studies of the anharmonic effects in the C₂H₄ + OH reaction suggest that only the anharmonicities arising from the lowest frequency torsional mode(s) need to be accounted for.⁷ One-dimensional corrections for such modes are easily incorporated.

Multi-reference second order perturbation theory (CASPT2)⁹⁷ provides an optimum approach for determining the rovibrational properties of the inner transition state. Such calculations combine high accuracy with suitable efficiency, particularly now that analytic CASPT2 gradients are available in electronic structure programs such as MOLPRO.⁹⁸

Furthermore, they allow for the consideration of multiple electronic states, which can be important for cases where the radical has degenerate electronic orbitals, as in $^2\Pi$ radicals. Spin orbit interactions can also be explicitly incorporated.

The use of an augmented polarized valence triple zeta basis set, such as that of Dunning and coworkers,⁹⁹ generally provides satisfactorily converged geometries and vibrational frequencies. The related augmented double zeta basis set, aug-cc-pvdz, is also often sufficient for these purposes. However, for the latter basis set significant problems have occasionally been noted, particularly for unsaturated species.^{9,100}

Coupled cluster calculations provide an alternative high level approach for performing the rovibrational analysis. When perturbative corrections for triples contributions are included (i.e., with the CCSD(T)¹⁰¹ or QCISD(T)¹⁰² methods), such analyses are typically expected to be of somewhat higher accuracy than CASPT2 based analyses. The T1 diagnostic¹⁰³ provides a useful measure of the multi-reference character to the wavefunction. The reliability of CCSD(T) calculations decreases dramatically for T1 values beyond 0.03. Unfortunately, coupled cluster calculations are considerably more intensive computationally. Typically, the difference between the CASPT2 and CCSD(T) predicted geometries and vibrational frequencies is negligible, and so the additional expense is generally not warranted. Occasionally, one is tempted to implement a CCSD(T) analysis, but with a smaller basis set. However, the errors introduced by basis set limitations are often greater than those arising from the limitations of the CASPT2 approach. The CCSD and QCISD approaches are also sometimes quite effective. However, it is not clear whether such analyses are generally an improvement over CASPT2 analyses.

The more limited geometry and vibrational analyses employed in standard *ab initio* thermodynamics schemes, such as G3,¹⁰⁴ while typically suitable for thermodynamic evaluations of stable species, lack the accuracy required for quantitative *a priori* kinetic predictions. In essence, kinetic predictions depend much more strongly on the vibrational analyses than do low temperature thermodynamic predictions. Furthermore, saddle point geometries are more strongly dependent on the electronic structure methodology than are equilibrium geometries.

Some limitations in such MP2 or density functional theory analyses can be accounted for by mapping out the reaction path and then using a higher level method to locate a maximum along the path. The projected frequencies along the low level reaction path can be used to obtain a variational transition state theory treatment. For low temperatures, the true variational effects are generally negligible, but the improved estimate of the high level saddlepoint (and its properties) can make a significant correction to the rate estimates.

Once the saddlepoint has been located it is useful to perform higher level estimates of its energies with methods such as CCSD(T) or QCISD(T). For such calculations, it is important to use an even larger basis, such as the aug-cc-pvqz basis. The results from calculations with triple and quadruple zeta basis sets can be extrapolated to yield improved predictions.¹⁰⁵ Notably, the errors arising from limitations to the triple zeta basis set can be rather large when considering second row atoms such as S (e.g., 15 kcal/mol). When CCSD(T) calculations with a quadruple zeta basis are not feasible, one can instead append basis set corrections obtained from MP2 or CASPT2 simulations. Other higher level schemes such as HEAT,¹⁰⁶ Wn,¹⁰⁷ or Gn,¹⁰⁸ provide alternative procedures for estimating the saddle point energy.

Notably, sample calculations with the CASPT2 method suggest that its predictions for the saddlepoint energy are remarkably accurate when the saddlepoint is submerged or nearly so. Indeed, the CASPT2 predictions appear to be even more accurate than CCSD(T) predictions. More generally, when considering energies for chemically dissimilar structures, the CCSD(T) approach is considerably more accurate than the CASPT2 approach. The efficacy of the CASPT2 approach for the inner transition state region of rapid low temperature reactions appears to be related to the relatively modest change in their bonding from the reactants to the inner transition state. Regardless, CASPT2 has become our method of choice for treating both the energies and the rovibrational properties of low temperature radical-molecule kinetics. Comparisons with multi-reference configuration interaction and QCISD(T) calculations can be used to confirm the accuracy of the CASPT2 calculations. When differences are noted, an expansion of the active space can sometimes be used to resolve the discrepancy.

4.3.1.2. Outer Transition State

Ab initio electronic structure simulations are also useful when considering the long-range transition state. The direct incorporation of *ab initio* simulations into the evaluation of the long-range transition state theory capture rate according to Eq. (4.10) removes any uncertainties in the appropriateness of an analytic representation for the long-range potential. The appropriateness of the underlying *ab initio* methods can be tested by comparing the predicted and experimental electrostatic properties such as the dipole and quadrupole moments and polarizabilities. Such comparisons suggest that the aug-cc-pvdz basis is generally large enough to provide converged long-range potentials, although the aug-cc-pvtz potential is preferred.

Typically, the long-range potential is not especially sensitive to the optimized geometry. Furthermore, there is essentially no variation in the

optimized geometry of the reacting fragments in the long-range region of the potential (i.e., for separations of 5 Å or greater.) Thus, relatively low level quantum chemical methods may be employed in the geometry optimizations that precede the evaluation of the long-range transition state number of states according to Eq. (4.10).

Importantly, *ab initio* simulations of the long-range transition state must employ a method that incorporates dynamic correlation. Otherwise the dispersion interaction, which is generally a dominant term in the expansion, will be neglected. Thus, one cannot employ methods such as HF, CASSCF or traditional DFT methods. (Note, however, the recent progress in developing DFT methods for treating dispersion.¹⁰⁹)

MP2 simulations are generally of sufficient accuracy for the long-range region, and are efficient enough for the direct sampling based evaluation of Eq. (4.10). Sample calculations suggest that CCSD(T) methods yield only minor revisions in the predicted long-range capture rate. However, orbital degeneracies in the radical can lead to wavefunction convergence problems. In such cases, it is best to employ CASPT2 methods, with state averaging of the degenerate orbital configurations. Such CASPT2 simulations are again efficient enough to incorporate in a direct sampling approach.

4.3.2. Radical-Radical Systems

Standard single reference methods, including MP2, CCSD(T), and even DFT methods such as B3LYP, are not applicable to the treatment of radical-radical reactions. The closed shell, spin-restricted, versions of these approaches have singularities precisely in the transition state region. Spin-unrestricted versions yield qualitatively correct reaction path potentials, but the energies are not quantitatively correct. The resulting errors in the predicted kinetics generally exceed a factor of two.

For radical-radical reactions, the full mode coupling and anharmonicity effects for the relative and overall rotational motions must be explicitly accounted for. We have derived a direct variable reaction coordinate transition state theory approach that appears to yield accurate rate coefficients for a number of alkyl radical reactions.^{110,111} This approach is analogous to that embodied in Eq. (4.10) for the long-range transition state, but includes variational optimizations of the form of the reaction coordinate and does not make the large orbital moment of inertia assumption. A detailed description of this approach was provided in some of our recent articles.

In principle, the variational minimization of the form of the reaction coordinate, which is defined in terms of a fixed distance between pivot points for each of the reactants, allows for the treatment of the dynamical non-adiabaticities that are neglected in standard adiabatic channel models.

Indeed, the reaction coordinate optimization typically yields a reduction in the predicted rate coefficient by about a factor of two from that obtained for a center-of-mass reaction coordinate. The quantum adiabatic channel calculations are generally restricted to an assumed center-of-mass separation for the reaction coordinate.

Our applications of the variable reaction coordinate (VRC) TST approach strongly suggest that the CASPT2 method generally yields quantitatively accurate interaction energies for both the inner and outer transition state regions. In particular, careful comparisons with MRCI and full-CI calculations indicate only kinetically insignificant discrepancies in the predicted interaction energies. Furthermore, the kinetic predictions are generally in quantitative agreement with experiment. Nevertheless, we have occasionally noted discrepancies between MRCI and CASPT2 based kinetic predictions. In this instance, a study of the dependence on the number of orbitals included in the active space often resolves the discrepancy.

For saturated alkyl radical reactions, a simple 2 electron, 2 orbital active space is sufficient. For more complex radicals, such as resonantly stabilized radicals, it is important to include additional orbitals, such as the full set of Π orbitals, in the active space.¹¹² In the case of electronically degenerate radicals, it is again generally necessary to employ state averaged wavefunctions.

Typically, these applications have employed the cc-pvdz basis set to examine the orientation dependence, coupled with an aug-cc-pvtz analysis of 1-dimensional corrections along the reaction path potential. Sample aug-cc-pvqz calculations suggest that the latter basis set provides an essentially converged interaction potential. However, our recent analysis of the $\text{CH}_3 + \text{OH}$ reaction¹¹³ suggests that an aug-cc-pvdz basis is preferred for the orientational sampling when hydrogen bonding interactions are present.

4.4. Short-Range Interactions and the Two Transition State Model

There is a wide variation in the temperature dependence of neutral reactions that are rapid at low temperature. As noted by Smith *et al.*,⁷² the reactions of CN with C_2H_2 , NH_3 , and C_2H_6 , illustrate the range of behavior. For C_2H_2 , the rate coefficient is close to the long-range capture rate, and is nearly temperature independent; varying from 5×10^{-10} to $2 \times 10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ as the temperature increases from 23 to 700 K. In contrast, the rate coefficient for the reaction of C_2H_6 with CN takes a minimum value of $2 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at about 200 K, rising to $1 \times 10^{-10} \text{ cm}^3$

molecule⁻¹ s⁻¹ as the temperature falls to 23 K or rises to 1000 K. Notably, this low temperature value is still a factor of 4 below the long-range capture rate. The temperature dependence for the reaction with NH₃ is between these two extremes, with a rate equal to the capture rate at low temperature, but decreasing by a factor of 20 from 20 to 200 K.

Many neutral reactions have similar discrepancies between the long-range capture rate predictions and the observations from CRESU experiments. For example, only 14 of the long-range capture rate predictions from Ref. 4 are within a factor of two of the 26 CRESU results at their lowest temperature. (Note, however, that the comparison in Ref. 4 did not consider atomic reactants, which do tend to have rates that are closer to the capture rate.) Furthermore, even where there is agreement at the lowest temperature, the CRESU results often decrease with temperature, whereas the long-range capture predictions are typically nearly constant or increasing with temperature. In every instance of discordance, the capture theory predictions exceed the experimental observations. These discrepancies are suggestive of some alternative bottleneck to the reaction.

The formation of a chemical bond requires the orientation of the reactants in order to maximize the overlap of the electronic orbitals involved in the incipient bond. The restriction in the range of reactive orientations, and corresponding decrease in the entropy, becomes increasingly severe as the separation distance decreases. Meanwhile, the strength of the chemical bond increases with decreasing separation. The interplay between these two opposing contributions typically yields a maximum in the free energy, i.e., a transition state, at separations where the chemical bonding and long-range interactions are of about the same magnitude. This generally occurs at separations of about 2–4 Å, where both interactions are on the order of 1 kcal/mol.

Importantly, the orientations required for the incipient bond are generally quite different from the optimal orientation for the long-range interactions. As a result, it is fairly common for there to be a long-range van der Waals potential minimum that precedes the inner transition state. In this case, there are clearly two transition states along the path from reactants to chemically bound complex. A schematic illustration of these two transition states is provided in Fig. 4.5. One describes the formation of the van der Waals complex from reactants. Long-range capture theory should provide a good description of this outer transition state. The other transition state describes the formation of a chemical complex from the van der Waals molecule. The properties of this inner transition state arise from a delicate balance between bonding, short-range repulsion, and long-range interactions. As a result, the accurate treatment of this inner transition state requires detailed *ab initio* simulations.

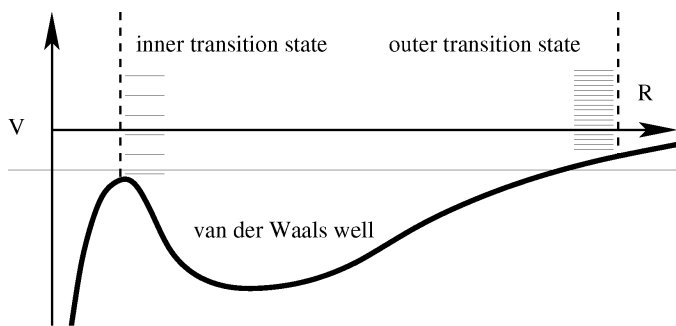


Fig. 4.5. Schematic diagram of the inner and outer transition states on a potential energy surface. The horizontal lines represent the energy levels at the two transition states.

One key result of the generally distinct nature of the inner and outer transition state regions is that long-range potential based predictions for the location of the outer transition state can underestimate the importance of the inner transition state. Such predictions only describe when the outer transition state reaches separations for which the long-range expansion is no longer appropriate. However, since its properties are distinct from those of the outer transition state, the inner transition state can provide the dominant bottleneck at much lower temperatures. Consider, for example, the limiting case where the energy of the inner transition state lies above the reactants, but there is still a well defined van der Waals complex. The inner transition state is then always the dominant bottleneck and the temperature at which the long-range transition state merges with the short-range transition state is irrelevant.

There are two factors that determine the relative importance of short-range interactions: (i) the energy of the inner transition state relative to asymptotic fragments and (ii) the entropy change at the inner transition state. The decrease in the entropy at the transition state is related to the change from free rotations of the fragments to nearly harmonic bends at the inner transition state for a number of the fragment rotational modes. Thus, the larger the number of rotational modes in the fragments, the greater the decrease in the entropy. For this reason, and due to the increased steric repulsion, the inner transition state is expected to be most important when considering the reaction of two nonlinear polyatomics and least important when considering the reaction of an atom with a diatom. The energy of the inner transition state depends on the type of interaction: ionic, radical-molecule or radical-radical. We consider these three cases separately in the following subsections.

4.4.1. Radical-Molecule Reactions

For radical-molecule reactions the formation of a bond between the radical and the molecule is coupled with the breaking of some bond in one of the reactants. The need to break a bond of the reactants prior to reaction generally leads to a well defined saddlepoint in the potential energy along the reaction path. The height of this barrier relative to reactants is a key determining factor in the rate at low temperature. If the zero-point corrected barrier is above the reactants zero-point energy, as is most often the case, then the reaction will be slow at low temperature. As noted by Herbst and coworkers,^{114,115} with tunneling of low mass species such as H_2 , the low temperature rate may be significant, but still a few orders of magnitude below the collision rate. In contrast, if the barrier is “submerged” (i.e., lies below the reactants) then the low temperature limit should be equal to the long-range capture rate.

As recently pointed out by Smith *et al.*,⁷² for addition reactions of unsaturated hydrocarbons, the height of the barrier should correlate with the difference between the ionization energy (IE) of the unsaturated hydrocarbon and the electron affinity (EA) of the radical. They went on to note that simple arguments, as well as the experimental evidence, suggest that if this energy difference exceeds 8.75 eV then the barrier should be above the reactant asymptote. Conversely, if it is less than 8.75 eV then the barrier should be submerged and the reaction should be rapid at low temperature.

A subsequent combined experimental and theoretical study of the reaction of $O(^3P)$ atoms with a variety of alkenes provided further confirmation for this correlation.⁹ The minimum energy path potentials for the two reactive surfaces in the $O(^3P) +$ ethylene and $O(^3P) +$ trans-butene reactions, cf. Fig. 4.6, clearly show the two transition state behavior for these reactions. For ethylene, the IE – EA difference is 9.05 eV and both barriers are significantly above the reactants’ energy. In contrast, for trans-butene this difference is 7.64 eV, and the ground state surface is submerged by more than 1 kcal/mol.

When this saddlepoint is submerged the long-range transition state will be the dominant bottleneck at the lowest energies and the reaction will be rapid at low temperatures. However, the energy levels for the intermolecular motions of the fragments are more widely spaced at the inner transition state than at the outer transition state (cf., Fig. 4.5). As a result, as the excess energy increases, the number of states at the inner transition state gradually becomes smaller than the number of states at the outer transition state. Thus, the inner transition state gradually becomes the dominant bottleneck as the energy increases.

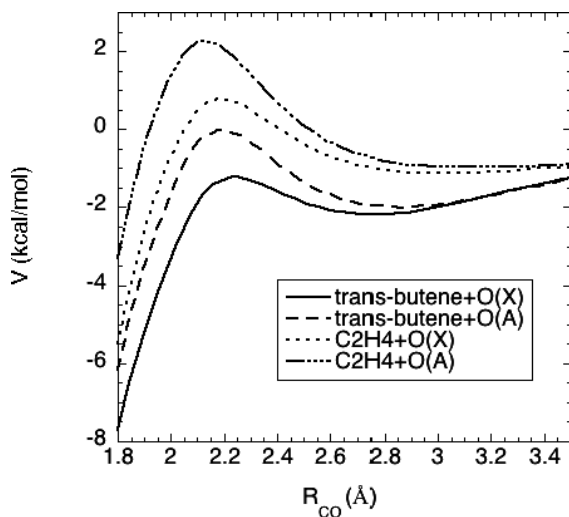


Fig. 4.6. Plot of the potential energy along the minimum energy path for the ground and first excited states of $O(^3P)$ reacting with ethylene and with trans-butene.

For some intermediate energy the numbers of states at the two transition states, $N_{\text{inner}}^\ddagger$ and $N_{\text{outer}}^\ddagger$, are roughly equivalent. In this case, the two transition states are acting in series and yield a greater reduction in the flux than would be expected from either one individually. The simple expression

$$1/N_{\text{eff}}^\ddagger = 1/N_{\text{inner}}^\ddagger + 1/N_{\text{outer}}^\ddagger \quad (4.23)$$

provides an accurate treatment of the net reactive flux if the dynamics is statistical and the well between the two transition states is reasonably deep. Alternatively, the unified statistical theory expression

$$1/N_{\text{eff}}^\ddagger = 1/N_{\text{inner}}^\ddagger + 1/N_{\text{outer}}^\ddagger - 1/N_{\text{max}} \quad (4.24)$$

is more appropriate if the well is shallow enough that the flux through it, N_{max} , is not much greater than that through the two transition states.¹¹⁶ Note that Eq. (4.24) reduces to Eq. (4.23) in the limit of large N_{max} .

A complete treatment of the kinetics through the van der Waals region should involve a master equation simulation¹¹⁷ of collision induced changes in the energy. However, simplifying assumptions are appropriate in the limit of low or high pressure. The appropriateness of these limits depends on both the pressure and the van der Waals well depth. If there are enough collisions to yield thermal equilibrium populations of the van der Waals energy levels, then the high pressure limit applies and the numbers of states should be replaced with canonical partition functions. If instead there are

no collisions, then the numbers of states should be evaluated for fixed total energy and total angular momentum, since those quantities are conserved between collisions. Our own sample master equation simulations suggest that, because of the small well depth for van der Waals complexes, the collisionless limit, where one employs E and J resolved numbers of states, is almost always appropriate.

A two transition state model for radical molecule reactions was postulated long ago by Singleton and Cvetanovic in their analysis of the reaction of $O(^3P)$ atoms with olefins.¹¹⁸ Unfortunately, their analysis focused on the canonical implementation and numerous subsequent *ab initio* implementations of the two transition state model have followed their lead. The canonical implementation can yield as much as a factor of 6 overestimate of the rate coefficient for temperatures near the crossover temperature (the temperature at which the inner and outer rate coefficients are equivalent), as illustrated in Fig. 4.7 for the $OH + C_2H_4$ reaction. Notably, the E/J resolved two transition state model predictions deviate significantly from the single transition state results over the full temperature range from 20 to 300 K.

An *ab initio* based variational TST study demonstrated that short-range interactions were central to the low temperature kinetics of the $CN + O_2$ radical-radical reaction⁶ as suggested in an earlier study of Clary and coworkers.⁴³ This result implied that short-range interactions could be of

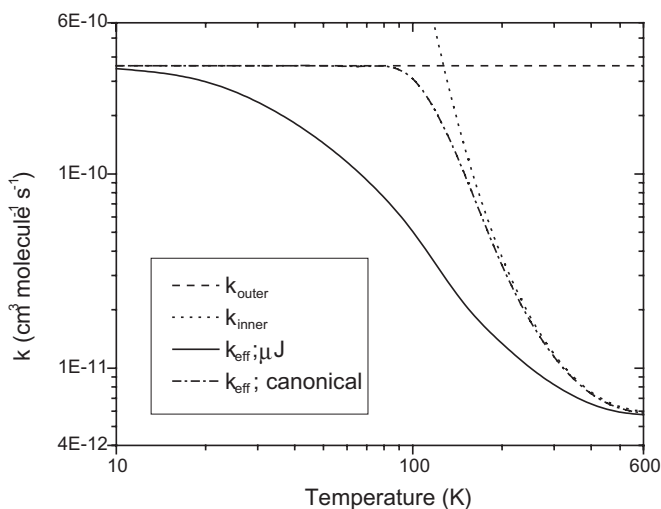


Fig. 4.7. Plot of the inner, outer, and the canonical and E/J resolved implementations of the effective two transition state model predictions for the $OH + C_2H_4 \rightarrow C_2H_4OH$ reaction.

similar importance in radical-molecule reactions. Clary and coworkers suggested their importance as an explanation for the failure of their long-range potential based predictions to reproduce experiment for the $\text{OH} + \text{HBr}$,¹¹⁹ $\text{CN} + \text{NH}_3$, and $\text{CH} + \text{NH}_3$ reactions.⁴⁵ Valiron and coworkers developed a simple model of rotational selectivity arising from the inner transition state that they applied to the $\text{CN} + \text{NH}_3$ reaction and to an illustrative atom-diatom system.^{120,121}

Herbst and coworkers suggested the importance of short-range interactions for the $\text{CN} + \text{C}_2\text{H}_2$ ³ and the $\text{C}_2\text{H} + \text{C}_2\text{H}_2$ reactions.¹²² For the latter reaction, they estimated the range of validity of the long-range capture rate simply on the basis of the temperature dependence of the long-range centrifugal barriers. As discussed in the Section 4.4 overview, this estimate is unlikely to be correct, and indeed their long-range predictions, which they claim should be valid up to 50 K, were a factor of two higher than the subsequent experimental results⁹³ all the way down to 15 K. For the former reaction, Woon and Herbst subsequently demonstrated that even minor deviations from the long-range capture rate are not necessarily indicative of an inner transition state for the entrance channel.⁵ In particular, a barely submerged exit channel transition state has a similar effect to one in the entrance channel as long as the pressure is not high enough to stabilize the molecular complex.

Both Clary and coworkers¹²³ and Herbst and coworkers³ have commented on the importance of short range interactions for the reaction of $\text{C}(^3\text{P})$ with acetylene. As Liao and Herbst pointed out, the presence of only one reactive potential surface at short-range greatly improves the agreement with experiment. Buonomo and Clary's subsequent reduced dimensionality quantum dynamics study¹²⁴ found good agreement with experiment when employing a global potential energy surface. However, this study suggested that ACCSA predictions overestimate the rate by a factor of two, due to significant non-adiabaticities in the dynamics.

The two transition state model was first applied to a rapid radical-molecule reaction at the E/J resolved level by Rauk and coworkers in their study of the reaction of CH_3 with HBr .¹²⁵ Related two transition state models were considered by Jodkowski *et al.*, for a number of abstraction reactions with CH_3OH ¹²⁶ and by Lin and coworkers in their study of the $\text{OH} + \text{H}_2\text{CO}$ reaction.¹²⁷ A simplified two transition state model, termed modified transition state theory, was recently introduced by Krasnoperov *et al.*¹²⁸ The latter model only considers the effect of the outer transition state on the threshold energies and so would not be expected to yield accurate low temperature rate coefficients. The other two transition state models also employ various simplified treatments, particularly for the outer transition state.

Our earliest *ab initio* implementation of the two transition state model to a barrierless reaction was to the dissociation of ketene into $^1\text{CH}_2 + \text{CO}$.¹²⁹ This reaction is an interesting case, since there is no saddle point on the minimum energy path, and yet photodissociation studies of the energy dependence of the dissociation rate and product rovibronic distributions strongly suggest an inner transition state. Our variational RRKM studies of this dissociation, which progressed with time from empirical model potentials¹³⁰ to high level *ab initio* studies,¹³¹ provided a quantitative description for these experimental observations in terms of an inner transition state that becomes the dominant transition state at an excess energy of about 140 cm^{-1} . The effects of this inner transition state are significant down to 50 cm^{-1} , and so would affect the reverse association rate constant down to 100 K or lower.

More recently, we have implemented the two transition state model for calculations of the $\text{SiH}_2 + \text{SiH}_4$,¹³² $\text{C}_2\text{H}_4 + \text{OH}$,⁷ $\text{CN} + \text{C}_2\text{H}_6$,⁸ $\text{OH} + \text{isoprene}$,¹³³ and $\text{O}(^3\text{P}) + \text{alkene}$ reactions.⁹ These implementations employ the long-range transition state theory treatment of the outer transition state and so are expected to provide an accurate treatment of the low temperature behavior. Furthermore, for some of the reactions, anharmonic and variational effects for the inner transition state were explicitly treated. For each of these reactions, the inner transition state was predicted to significantly decrease the rate constant in the low temperature regime, with all but the isoprene reaction having important effects down to 20 K.

The results from our *a priori* implementation of the two transition state model for the $\text{CN} + \text{C}_2\text{H}_6$ reaction are illustrated in Fig. 4.8. These results quantitatively reproduce the experimental observations from 25 to 1140 K.^{90,134–138} Minima in rate versus temperature plots are usually indicative of the opening of new reactive pathways. However, for this reaction, the minimum in the rate constant versus temperature is simply the result of the changing dominance of the inner and outer transition states.

Our predictions for the $\text{O}(^3\text{P}) + \text{alkenes}$ reaction demonstrate the changing behavior as the saddlepoint becomes increasingly submerged. As illustrated in Fig. 4.9, for the propene case, which has a slightly positive barrier of 0.1 kcal/mol, the rate coefficient is about two orders of magnitude below the long-range capture rate. In this case, the outer transition state plays almost no role in the kinetics. In contrast, the rate constant for trans-butene, whose barrier is submerged by 0.9 kcal/mol, is an order of magnitude greater than for propene. In this case, both transition states are important all the way from a few K to 200 K.

The $\text{O}(^3\text{P}) + \text{trans-butene}$ reaction also provides an illustration of the complications that arise from our imprecise knowledge of the electronic dynamics. In particular, consider the schematic correlation diagram of the

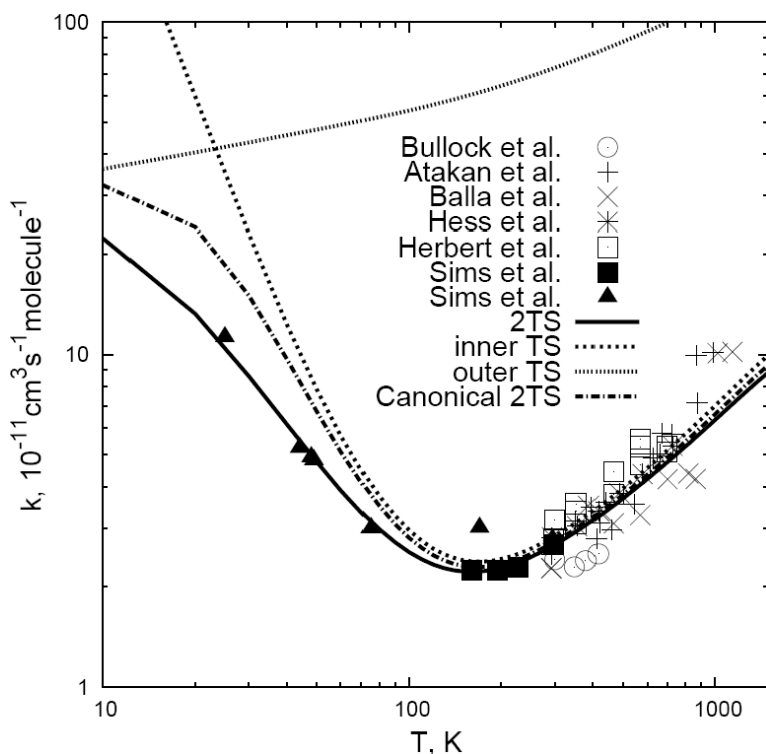


Fig. 4.8. Plot of the experimental and the long-range, short-range and two transition state model predictions for the temperature dependence of the rate constant for $\text{CN} + \text{C}_2\text{H}_6 \rightarrow \text{HCN} + \text{C}_2\text{H}_5$.

electronic states for this reaction shown in Fig. 4.10. The ground ($^3\text{P}_2$) state of the O atom is fivefold degenerate. At the inner transition state only the ground triply degenerate state (X) lies below reactants. The first excited state (A) is slightly positive relative to ground state reactants.

Thus, if the electronic dynamics is fully adiabatic (from reactants to the covalent molecular complex), one should employ an electronic degeneracy factor of 3 for both $N_{\text{inner}}^\ddagger$ and $N_{\text{outer}}^\ddagger$. Then, at low temperature, where the $^3\text{P}_1$ and $^3\text{P}_0$ states of the O atom are not occupied, the overall rate coefficient is predicted to be the simple long-range capture rate multiplied by 3/5. If instead, there is a statistical mixing of electronic states in the pre-reaction complex, then one should again employ an electronic degeneracy factor of 3 for $N_{\text{inner}}^\ddagger$, but the degeneracy factor for $N_{\text{outer}}^\ddagger$ should now be 5. In that case, the predicted low temperature rate is simply the long-range capture rate; i.e., the multiplicative factor is 5/5.

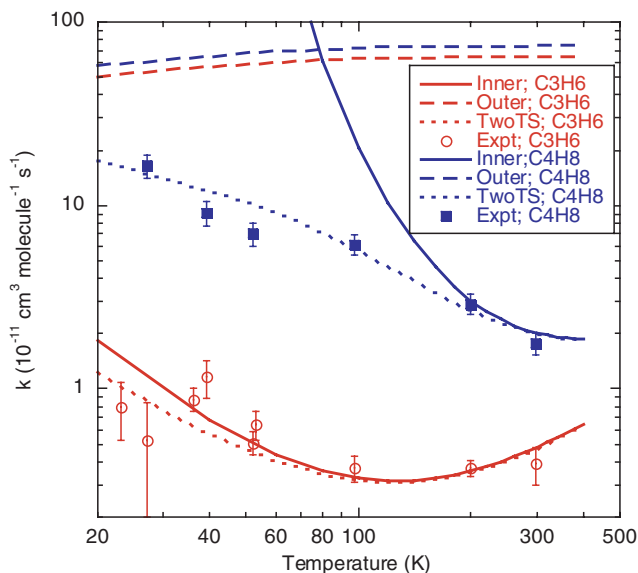


Fig. 4.9. Plot of the experimental and the long-range, short-range and two transition state model predictions for the temperature dependence of the rate constant for $\text{O}(^3\text{P}) + \text{propene}$ and $\text{O}(^3\text{P}) + \text{trans-butene}$ reactions.

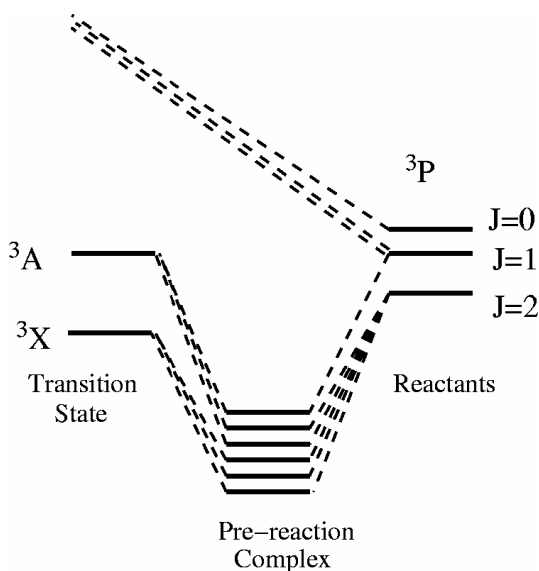


Fig. 4.10. Schematic plot of the correlation of the electronic states from reactants to the inner transition state for the $\text{O}(^3\text{P}) + \text{trans-butene}$ reaction.

The calculations from Ref. 9 employ the adiabatic electronic assumption. This assumption provides slightly improved agreement with experiment at the lowest temperatures observed (27 K). However, the distinction between the adiabatic and statistical predictions is still quite minor at 27 K, because the inner transition state still plays an important role. It would be useful to have some experimental results in the long-range capture dominated temperature regime for some system where the adiabatic statistical and electronic predictions differ.

For reactions such as $\text{OH} + \text{C}_2\text{H}_4$, the rate coefficient may differ from the long-range capture rate due to an absence of product channels with an exothermic pathway. In such simple addition reactions, the measured rate coefficient should be pressure dependent. For the $\text{OH} + \text{C}_2\text{H}_4$ reaction we have incorporated our two TS predictions in a master equation model for the pressure dependence of the kinetics.⁷ The barrier height was adjusted down by 1 kcal/mol from high level *ab initio* predictions in order to reproduce the room temperature high pressure rate data. As illustrated in Fig. 4.11, the resulting predictions for the full temperature and pressure dependence were in quantitative agreement with the experimental data,^{139–142} including that from Leone and coworkers in the 96 to 165 K region.

Recently, however, Blitz and coworkers have provided two additional low temperature experimental studies.^{143,144} Both of these studies suggest that

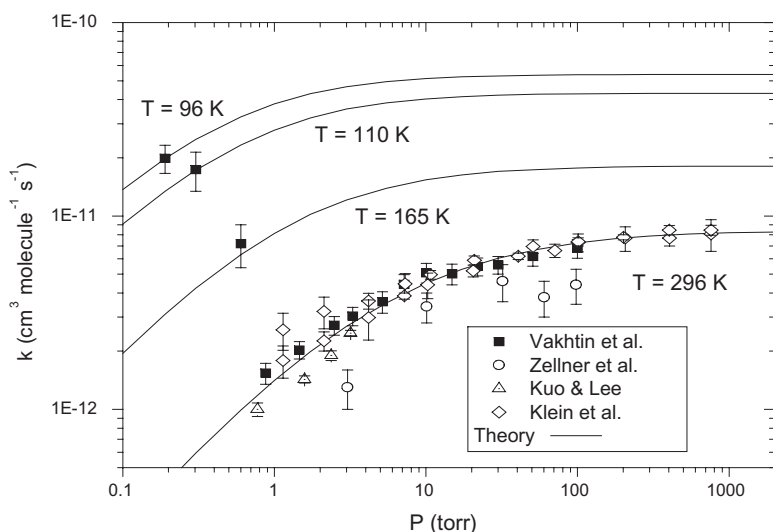


Fig. 4.11. Plot of the pressure dependence of the low temperature rate coefficients for $\text{OH} + \text{C}_2\text{H}_4 \rightarrow \text{C}_2\text{H}_4\text{OH}$.

our predictions for the high pressure limit are too high, apparently by about a factor of two to three in the 69 to 96 K region. Such a discrepancy may be indicative of a failure of the effective transition state formula, Eq. (4.23). It would be interesting to perform a dynamical study for this reaction to test the accuracy of Eqs. (4.23) and (4.24) for a given potential energy surface.

Blitz and coworkers have also modeled their data with master equation simulations employing empirical high pressure rate values. While these simulations do provide improved agreement with the full set of data, they do so by splitting the difference between theirs and Leone and coworkers experimental results. The remaining discrepancies between the fit and experiment are outside the estimated experimental error bars, suggesting some discrepancy between the various experiments. Thus, it would be useful to have further experimental studies of this reaction.

4.4.2. Ion-Molecule Reactions

Long-range complexes are ubiquitous in ion-molecule chemistry due to the increased strength of the long-range attractions. They tend to arise whenever the formation of a covalently bound complex requires structural rearrangements. With this increased attraction, the saddle points connecting the long-range ion-molecule complexes to the covalent compounds also tend to lie well below reactants. Thus, the inner transition state seldom plays an important role and, as discussed in Section 2, the long-range capture rates generally provide an accurate prediction of the kinetics for room temperature and lower. Nevertheless, if the long-range attractions are weak enough the inner transition state can be important at low temperatures even for ion-molecule reactions.

Bowers and coworkers were the first to consider two transition state models for ionic reactions. In their classic study of the decomposition of C_4H_8^+ cations,¹⁴⁵ they considered two transition state models for each of the three product channels, $\text{C}_2\text{H}_4 + \text{C}_2\text{H}_4^+$, $\text{H} + \text{C}_4\text{H}_7^+$, and $\text{CH}_3 + \text{C}_4\text{H}_5^+$. In subsequent work, they applied a related model to the dissociations of C_4H_6^+ ¹⁴⁶ and C_6H_6^+ .¹⁴⁷ The implementations for the inner transition state were purely empirical, but did provide clear evidence for its kinetic importance.

With its low polarizability and nonpolar nature, the H atom has the weakest possible long-range interaction with ions. Furthermore, its small mass implies that the centrifugal barriers are relatively large and lie at shorter separations. Correspondingly, one expects short range interactions to be of more importance for H^+ ion reactions than for other ion-molecule reactions. We have studied the $\text{H} + {}^3\text{C}_6\text{H}_5^+$ recombination reaction with direct CASPT2/aug-cc-pvdz VRC-TST calculations employing a 7 electron,

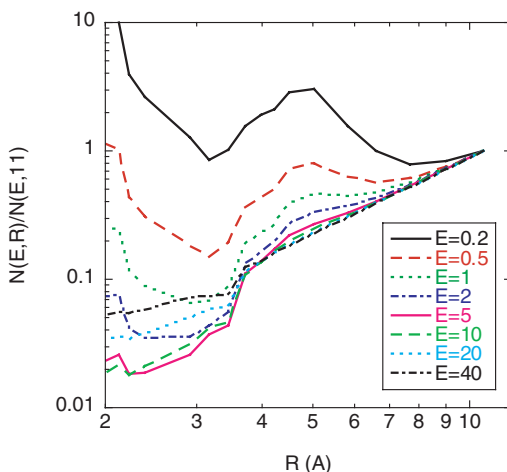


Fig. 4.12. Plot of the normalized microcanonical number of states as a function of separation distance for a range of energies (kcal/mol) for $\text{H} + {}^3\text{C}_6\text{H}_5^+ \rightarrow \text{C}_6\text{H}_6^+$. The numbers are normalized to their value at $R = 11 \text{ \AA}$.

8 orbital active space consisting of the H atom radical orbital, the C_6H_5^+ in-plane radical orbital and the C_6H_5^+ Π space. At long range, $R > 4 \text{ \AA}$, the reaction coordinate was taken to be the center-of-mass separation distance. At short range, $r_{\text{CH}} < 4.2 \text{ \AA}$, a multifaceted dividing surface¹⁴⁸ consisting of fixed H to C atom separations was considered.

The radial dependence of the microcanonical number of states for the $\text{H} + {}^3\text{C}_6\text{H}_5^+ \rightarrow \text{C}_6\text{H}_6^+$ reaction over a range of energies is illustrated in Fig. 4.12. At the lowest energy plotted, 0.2 kcal/mol, an inner transition state, at $R \sim 3 \text{ \AA}$, and an outer transition state, at $R \sim 8 \text{ \AA}$, are clearly visible. At $E = 0.2 \text{ kcal/mol}$ these bottlenecks are essentially identical, while the inner bottleneck becomes dominant (i.e., has the smaller number of states) as the energy increases. For this reaction, the inner transition state would clearly play a significant role in the low temperature kinetics.

The methyl radical has a significantly greater polarizability than the H atom, but still has no dipole. Furthermore, there are a large number of transitional modes that transform from rotations to vibrations. The radial dependence of similar direct MP2/cc-pvdz VRC-TST calculations of the microcanonical number of states for the $\text{CH}_3 + \text{C}_3\text{H}_5^+ \rightarrow \text{C}_4\text{H}_8^+$ reaction are plotted in Fig. 4.13. The qualitative behavior is similar to that for the $\text{H} + {}^3\text{C}_6\text{H}_5^+$ reaction. But now $N_{\text{inner}}^\ddagger > N_{\text{outer}}^\ddagger$ until an energy of 2 kcal/mol is reached. In this case, the low temperature kinetics will be dominated by the long-range potential, but the inner transition state should still yield a modest reduction in the rate coefficient.

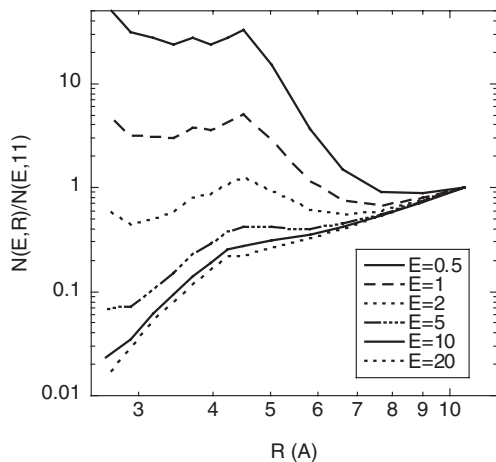


Fig. 4.13. Plot of the microcanonical number of states as a function of separation distance for a range of energies (kcal/mol) for $\text{CH}_3 + \text{C}_3\text{H}_5^+ \rightarrow \text{C}_4\text{H}_8^+$. The numbers are normalized to their value at $R = 11 \text{ \AA}$.

Ammonia provides a simple example of a molecule with a strong dipole and corresponding strong long-range interactions. However, since it is a closed shell, one might expect there to be a significant saddle point preceding its reaction. Direct MP2/cc-pvdz VRC-TST calculations for the reaction of NH_3 with C_3H_7^+ find that the inner and outer bottlenecks do not become equivalent until an energy of 8 kcal/mol. One would not expect to see any effect of short range interactions on the low temperature kinetics for this reaction.

Since most molecules do have a permanent dipole moment, the $\text{NH}_3 + \text{C}_3\text{H}_7^+$ case should be fairly typical. Unless the saddlepoint for reaction happens to lie close to or above the reactant energy, one does not expect significant effects from short-range interactions for the reaction of dipolar molecules and radicals with ions. For nonpolar radicals reacting with closed shell ions, short range interactions may be of modest significance. For closed shell ions reacting with nonpolar closed shell molecules or with H atoms, an inner transition state will generally have an effect on the low temperature kinetics. If the ion is a radical, the probability of an important inner transition state is even further reduced for each of these cases.

4.4.3. Radical-Radical Reactions

Neutral radical-radical reactions are quite different from radical-molecule reactions in that the formation of the incipient chemical bond does not require the breaking of some other covalent bond. Importantly, however,

the reactants still must rotate from the optimal long-range orientations to the optimal covalent orientations and there is a strong restriction in the range of accessible angles. As a result, an inner transition state can and generally does arise. In some cases this inner transition state correlates with a submerged saddle point on the potential energy surface, while in other cases it is solely due to the reduction in entropy. Furthermore, in some cases the inner and outer transition state regions are clearly distinct, while in others the location of the transition state just gradually shifts from large separations to short separations with no discernible separation into an inner and outer transition state.

The presence or absence of a saddlepoint and the distinction between inner and outer transition states depends subtly on various factors such as (i) the presence of any dipole moments, (ii) the presence of any hydrogen bonding interactions between the two reactants, (iii) the resonance stabilization of either reactant, (iv) the relative orientation of the optimal short-range and long-range interactions, (v) the number of rotational degrees of freedom of the reactants, and (vi) the molecular size of the reactants.

A model potential based variational RRKM study of the $\text{NCNO} \rightarrow \text{NC} + \text{NO}$ dissociation provided one of the first indications of the relevance of the two transition state model to the kinetics of radical-radical reactions.¹⁴⁹ In this study, the dynamics on the singlet and triplet surfaces at large range was assumed to be statistical in keeping with the observed statistical distribution of the rovibronic states of the products. The resulting larger degeneracy of the long-range region yielded a rapid transition to a short-range singly electronically degenerate inner transition state with an energy of 100 cm^{-1} .⁶⁷ By microscopic reversibility, this analysis is directly applicable to the reverse association and short-range effects should be dominant at low temperature. A subsequent experimental study of the $\text{CN} + \text{NO}$ association yielded an estimated high pressure rate constant of $\sim 5 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ over the 200 to 600 K range, which is in good agreement with that from the theoretical model.¹⁵⁰ By comparison, the long-range TST prediction for the dominant dispersion term is $\sim 7 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ for this same range. This long-range prediction assumes electronically adiabatic dynamics with reaction on only the ground singlet state, which yields an electronic correction factor of $4/(4 + 4 \exp(-179/T(\text{K})))$.

The $\text{CH}_3 + \text{H}$ and $\text{CH}_3 + \text{CH}_3$ reactions are two prototypical radical-radical reactions. The latter reaction is of some interest to the modeling of low temperature planetary atmospheres.¹⁵¹ For the $\text{CH}_3 + \text{H}$ reaction, the long-range interactions are much weaker and the reduced mass is smaller. Both of these factors suggest that short-range interactions should be more important for the H atom case. However, for the $\text{CH}_3 + \text{CH}_3$ case, there are three more modes that transform from free rotations to bending motions

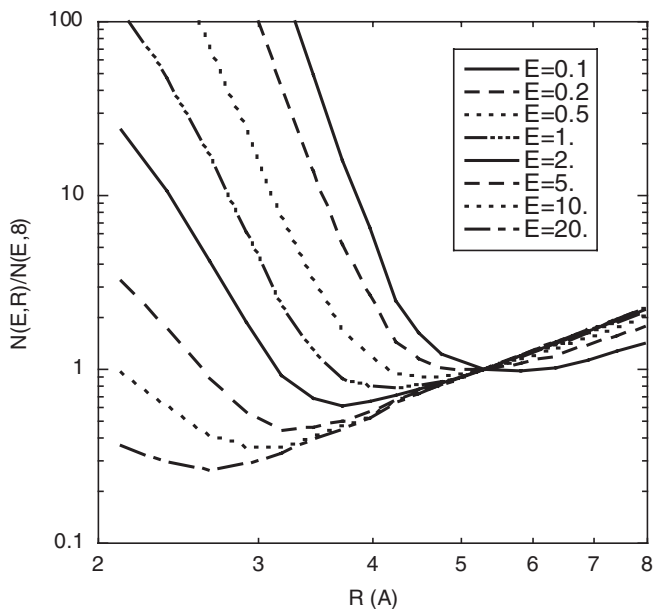


Fig. 4.14. Plot of the microcanonical number of states as a function of separation distance for a range of energies (kcal/mol) for $\text{CH}_3 + \text{H} \rightarrow \text{CH}_4$ reaction. The numbers are normalized to their value at $R = 8 \text{ \AA}$.

as the separation distance decreases. Furthermore, the additional molecular size yields more steric repulsion. These two factors yield a greater reduction in entropy, thereby suggesting that short range interactions may be more important in the $\text{CH}_3 + \text{CH}_3$ reaction.

The radial dependence of the direct CASPT2 calculated microcanonical number of states for these two reactions is plotted in Figs. 4.14 and 4.15, respectively. For the H atom case, the transition state gradually shifts from large separation to short separation, reaching a separation of 4–5 Å by an energy of 0.5 kcal/mol. There is no sign of two distinct transition state regions even if one plots the E/J resolved number of states. In contrast, the $\text{CH}_3 + \text{CH}_3$ case shows some indication of distinct outer and inner transition states, but the inner transition state only becomes significant at an energy of 2 kcal/mol. Furthermore, in this case, the distinct nature of the two regions is amplified if one instead considers the E/J resolved number of states.

The CASPT2/aug-cc-pvtz VRC-TST predicted rate coefficients for the $\text{CH}_3 + \text{H}$ and $\text{CH}_3 + \text{CH}_3$ reactions over the 10 to 1000 K region are plotted in Fig. 4.16. These predictions have previously been shown to be in good

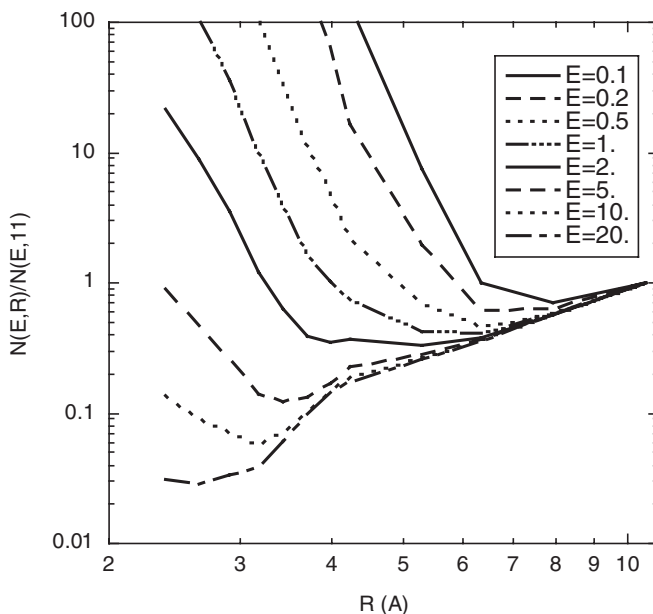


Fig. 4.15. Plot of the microcanonical number of states as a function of separation distance for a range of energies (kcal/mol) for $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$ reaction. The numbers are normalized to their value at $R = 11 \text{ \AA}$.

agreement with the available experimental data.^{110,111} Long-range TST predictions are also plotted for a potential that includes only dispersion forces.

The $\text{CH}_3 + \text{H}$ predictions are remarkably similar to the long-range TST predictions. Results for H atom plus other larger alkyl radical reactions suggest that this similarity is solely the result of a coincidental cancellation of two opposing factors. In particular, the H+ alkyl radical reactions decrease by about a factor of two for each substitution of an H atom in CH_3 with a CH_3 group.¹¹⁰ The dispersion rate for these larger radicals should be the same or larger since the polarizability is increased, while the reduced mass is essentially constant. The predicted decrease is an indication of short-range repulsions reducing the rate coefficient. This same factor in the $\text{H} + \text{CH}_3$ case is smaller. Furthermore, the increase in the attractive interactions due to bonding acts to increase the rate coefficient. Since the transition state reaches separations of about $4 \text{ \AA}$ by 1 kcal/mol , the lack of any significant deviation from the dispersion rate, indicates that the bonding and repulsion factors must cancel in the $\text{CH}_3 + \text{H}$ case.

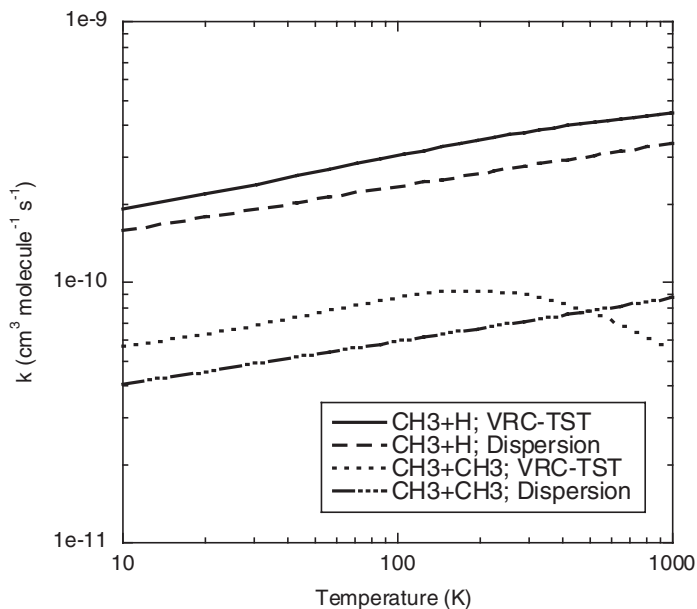


Fig. 4.16. Plot of the dispersion and VRC-TST predicted rate coefficients for $\text{CH}_3 + \text{H} \rightarrow \text{CH}_4$ and for $\text{CH}_3 + \text{CH}_3 \rightarrow \text{C}_2\text{H}_6$.

For the $\text{CH}_3 + \text{CH}_3$ reaction, the VRC-TST predictions begin to decrease with temperature at about 300 K. Below that temperature there is again good agreement with the long-range TST predictions for the dispersion potential. For larger alkyl radicals, the rate decreases with each additional CH_3 group,¹¹¹ which is again in contradiction with the expectations from long-range TST. Also, the temperature at which the rate begins to deviate from the long-range TST rate is predicted to decrease with increasing size.

Resonantly stabilized radicals, which are another important class of radicals, have much weaker attractions than do saturated alkyl radicals. This reduction in the attractiveness arises from the need to break the resonance as part of the bonding process. For these radicals short-range interactions yield a reduction in the rate coefficient at remarkably low temperatures. For example, for the $\text{CN} + \text{O}_2$ reaction, the resonances in the O_2 molecule must be broken prior to bonding.

The $\text{CN} + \text{O}_2$ reaction is one of the few radical-radical reactions that has been studied at low temperature with CRESU experiments.^{152–154} The observed rate coefficient for this reaction decreases sharply with increasing temperature above 23 K, whereas the long-range ACCSA capture rate prediction increases with temperature.⁴³ As illustrated in Fig. 4.17, an early

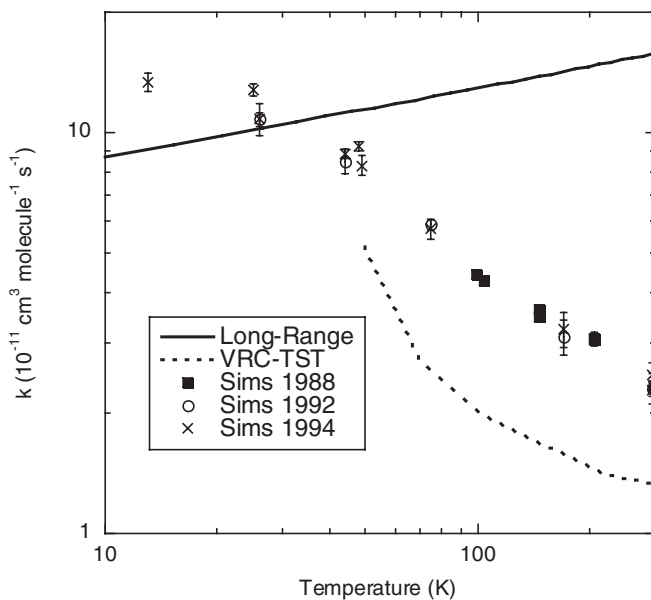


Fig. 4.17. Plot of experimental, long-range TST, and *ab initio* short-range transition theory results for the temperature dependence of the rate constant for $\text{CN} + \text{O}_2 \rightarrow \text{NCOO}$.

ab initio transition state theory based implementation of VRC-TST⁶ was able to describe this rapid decrease qualitatively on the basis of a short-range transition state. The discrepancies between these VRC-TST calculations and experiment can readily be explained by inadequacies in the treatment of the potential energy surface. The ACCSA predictions from Ref. 43 differ from the present long-range TST predictions, apparently by our inclusion of an electronic degeneracy factor of 1/3. This factor presumes that the reaction occurs on only the doublet electronic surface and not on the quartet surface. The underestimate at low temperatures for the long-range TST prediction suggests that either the quartet state does contribute some to the kinetics, or that the bonding and higher order interactions yield a modest increase in the long-range potential. Tan *et al.*'s study⁷⁶ of the contribution from different terms in the long-range potential, appears to have neglected the key dispersion term.

Other *ab initio* based theoretical studies of low temperature radical-radical kinetics have focused on atom-diatom reactions such as $\text{N} + \text{OH}$,¹⁵⁵ $\text{O} + \text{OH}$,¹⁵⁶ $\text{Si} + \text{O}_2$,¹⁵⁷ $\text{C} + \text{OH}$,¹⁵⁸ $\text{C} + \text{NO}$ ⁸⁰ and $\text{O} + \text{CN}$ ⁸⁰ reactions. The deviations from long-range predictions for these atom-diatom studies have generally been quite modest due to the small number of rotational degrees of

freedom. However, these studies do emphasize yet again, the strong dependence on the contributions from the multiple nearly degenerate electronic states.

Unfortunately, there have been only a few direct studies of the electronic dynamics. In a pioneering study, Tashiro and Schinke¹⁵⁹ examined the quantum dynamics of the $\text{O}(^3\text{P}) + \text{O}_2$ reaction on the multiple electronic surfaces arising from the spin-orbit states of the oxygen atom. In a related study Yagi *et al.*¹⁶⁰ studied the quantum dynamics of the $\text{CH}_3 + \text{O}(^3\text{P})$ reaction, but in only a two-dimensional framework. For both reactions, non-adiabatic effects were found to be of minimal importance. Nevertheless, it would be useful to examine these effects in other cases, particularly when statistical and adiabatic electronic treatments yield different estimates at low temperature.

4.5. Summary

In the low temperature limit the kinetics of gas phase reactions is determined by the capture rate on the long-range potential. The combination of statistical and dynamical, classical and quantum theories provides an accurate description of the long-range kinetics on a given potential energy surface. However, there are still significant uncertainties regarding the contributions from multiple degenerate electronic states. An assumption of electronically adiabatic dynamics generally yields good agreement with experiment and is the most commonly employed assumption.

Short-range interactions can yield a significant reduction in the predicted rate coefficient below that predicted by the long-range capture theory. One common effect of short-range interactions is to yield a change in the electronic degeneracy factors as only a limited number of potential energy surfaces are reactive at short-range. Another effect is to yield an inner transition state arising from a reduction in the entropy as the reactants approach one another. This inner transition state is often distinct from the outer or long-range transition state, especially for radical-molecule reactions. Two transition state models provide an effective treatment of this series of transition states. However, the accuracy of the simple models given by Eqs. (4.23) and (4.24) is still uncertain.

The short-range entropy reduction effect can become important already at 10 K, as seen in the study of the $\text{CN} + \text{C}_2\text{H}_6$ and $\text{O}(^3\text{P}) + \text{alkene}$ reactions. The accurate treatment of the short-range effects requires detailed and high level *ab initio* simulations. The CASPT2 method is a powerful approach for treating the short-range energetics in barrierless radical-radical and radical-molecule reactions. Modest adjustments in the *ab initio* predicted barrier

heights can generally be used to obtain theoretical predictions that are in quantitative agreement with experiment over a wide range of temperature.

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References

1. Smith IWM, Rowe BR. (2000) Reaction kinetics at very low temperatures: Laboratory studies and interstellar chemistry. *Acc. Chem. Res.* **33**: 261–268.
2. Smith IWM. (2006) Reactions at very low temperatures: Gas kinetics at a new frontier. *Angew. Chem. Int. Ed.* **45**: 2842–2861.
3. Liao Q, Herbst E. (1995) Capture calculations for the rates of important neutral-neutral reactions in dense interstellar clouds — $C + C_2H_2$ and $CN + C_2H_2$. *Astrophys. J.* **444**: 694–701.
4. Georgievskii Y, Klippenstein SJ. (2005) Long-range transition state theory. *J. Chem. Phys.* **122**: 194103; 1–15.
5. Woon DE, Herbst E. (1997) The rate of the reaction between CN and C_2H_2 at interstellar temperatures. *Astro. J.* **477**: 204–208.
6. Klippenstein SJ, Kim YW. (1993) Variational statistical study of the $CN + O_2$ reaction employing *ab-initio* determined properties for the transition state. *J. Chem. Phys.* **99**: 5790–5799.
7. Greenwald EE, North SW, Georgievskii Y, Klippenstein SJ. (2005) A two transition state model for radical-molecule reactions: A case study of the addition of OH to C_2H_4 . *J. Phys. Chem. A.* **109**: 6031–6044.
8. Georgievskii Y, Klippenstein SJ. (2007) Strange kinetics of the $C_2H_6 + CN$ reaction explained. *J. Phys. Chem. A.* **111**: 3802–3811.
9. Sabbah H, Biennier L, Sims IR, Georgievskii Y, Klippenstein SJ, Smith IWM. (2007) Understanding reactivity at very low temperatures: The reactions of oxygen atoms with alkenes. *Science* **317**: 102–105.
10. Clary DC. (1990) Fast chemical reactions: Theory challenges experiment. *Ann. Rev. Phys. Chem.* **41**: 61–90.
11. Troe J. (1997) Recent advances in statistical adiabatic channel calculations of state specific dissociation dynamics. *Adv. Chem. Phys.* **101**: 819–851.
12. Pechukas P, Light JC. (1965) On detailed balancing and statistical theories of chemical kinetics. *J. Chem. Phys.* **42**: 3281–3291; Pechukas P, Rankin R,

- Light JC. (1966) Statistical theory of chemical kinetics: Application to neutral-atom—molecule reactions. *J. Chem. Phys.* **44**: 794–805; Light JC. (1967) Statistical theory of bimolecular exchange reactions. *Faraday Discuss Chem. Soc.* **44**: 14–29.
13. Nikitin EE. (1965) *Teor. Eksp. Khim. Acad. Nauk. Ukr. SSR* **1**: 135, 428.
14. Langevin PM. (1905) *Ann. Chem. Phys.* **5**: 245; Gioumousis G, Stevenson DP. (1958) Reactions of gaseous molecule ions with gaseous molecules. V. Theory. *J. Chem. Phys.* **29**: 294–299.
15. Bass L, Su T, Bowers MT. (1978) Theory of ion-molecule collisions. Effect of anisotropy in polarizability on collision rate constant. *Int. J. Mass Spec. Ion Proc.* **28**: 389–399.
16. Clair R, McMahon TB. (1978) Orientation-dependent forces in ion-molecule reactions. Effect of anisotropic molecular polarizability on ion-molecule reaction-rates. *Int. J. Mass Spec. Ion Proc.* **28**: 365–375.
17. Hirschfelder JO, Curtiss CF, Bird RB. (1967) *Molecular Theory of Gases and Liquids*. John Wiley & Sons, New York.
18. Gorin E. (1939) Photolysis of aldehydes and ketones in the presence of iodine vapor. *J. Chem. Phys.* **7**: 256–264.
19. Nikitin EE, Troe J. (1997) Quantum and classical calculations of adiabatic and non-adiabatic capture rates for anisotropic interactions. *Ber. Bunsenges. Phys. Chem.* **101**: 445–458.
20. Benson SW. (1983) Molecular-models for recombination and disproportionation of radicals. *Can. J. Chem.* **61**: 881–887.
21. Smith GP, Golden DM. (1978) Application of RRKM theory to reactions $\text{OH} + \text{NO}_2 + \text{N}_2 \rightarrow \text{HONO}_2 + \text{N}_2(1)$ and $\text{ClO} + \text{NO}_2 + \text{N}_2 \rightarrow \text{ClONO}_2 + \text{N}_2(2)$: a modified gorin model transition state. *Int. J. Chem. Kinet.* **10**: 489–501.
22. Chesnavich WJ, Su T, Bowers MT. (1980) Collisions in a non-central field. Variational and trajectory investigation of ion-dipole capture. *J. Chem. Phys.* **72**: 2641–2655.
23. Su T, Bowers MT. (1973) Theory of ion-polar molecule collisions. Comparisons with experimental charge transfer reactions of rare gas ions to geometric isomers of difluorobenzene and dichloroethylene. *J. Chem. Phys.* **58**: 3027–3037.
24. Davidsson J, Nyman G. (1988) Extended Langevin model for rate constant calculations of exothermic atom-diatom reactions. Application to $\text{O} + \text{OH}$. *Chem. Phys.* **125**: 171–183.
25. Celli F, Weddle G, Ridge DP. (1980) On statistical and thermodynamic approaches to ion polar molecule collisions. *J. Chem. Phys.* **73**: 801–812.
26. Markovic N, Nordholm S. (1989) Simple estimation of thermal capture rates for ion-dipole collisions by canonical effective potential methods. *Chem. Phys.* **135**: 109–122.
27. Larregaray P, Bonnet L, Rayez JC. (2007) Mean potential phase space theory of chemical reactions. *J. Chem. Phys.* **127**: 084308; 1–4.
28. Turulski J, Niedzielski J. (1988) Implementation and limitations of the transition state theory for ion molecule systems with non-spherical dividing surfaces. *J. Chem. Soc. Faraday Trans.* **84**: 347–361.

29. Turulski J, Niedzielski J, Pezler B. (1995) The capture rate of an ion by a symmetrical top quadrupole. *Chem. Phys.* **192**: 319–323.
30. Smith SC, Troe J. (1992) Statistical modeling of ion-molecule electrostatic capture. *J. Chem. Phys.* **97**: 5451–5464.
31. Mondro SL, Vande Linde S, Hase WL. (1986) Reaction path and variational transition state theory rate constant for $\text{Li}^+ + \text{H}_2\text{O} \rightarrow \text{Li}^+(\text{H}_2\text{O})$ association. *J. Chem. Phys.* **84**: 3783–3787.
32. Wardlaw DM, Marcus RA. (1988) On the statistical theory of unimolecular processes. *Adv. Chem. Phys.* **70**: 231–263.
33. Su T. (1988) Trajectory calculations of ion-polar molecule capture rate constants at low-temperatures. *J. Chem. Phys.* **88**: 4102–4103; **89**: 5355–5355.
34. Bhowmik PK, Su, T. (1991) Parametrization of the trajectory calculations on the ion-quadrupolar molecule collision rate constants. *J. Chem. Phys.* **94**: 6444–6445.
35. Maergoiz AI, Nikitin EE, Troe J, Ushakov VG. (1996) Classical trajectory and adiabatic channel study of the transition from adiabatic to sudden capture dynamics. I. Ion-dipole capture. *J. Chem. Phys.* **105**: 6263–6269.
36. Maergoiz AI, Nikitin EE, Troe J, Ushakov VG. (1996) Classical trajectory and adiabatic channel study of the transition from adiabatic to sudden capture dynamics. II. Ion-quadrupole capture. *J. Chem. Phys.* **105**: 6270–6276.
37. Maergoiz AI, Nikitin EE, Troe J, Ushakov VG. (1996) Classical trajectory and adiabatic channel study of the transition from adiabatic to sudden capture dynamics. III. Dipole-dipole capture. *J. Chem. Phys.* **105**: 6277–6284.
38. Maergoiz AI, Nikitin EE, Troe J, Ushakov VG. (1998) Classical trajectory and statistical adiabatic channel study of the dynamics of capture and unimolecular bond fission. IV. Valence interactions between atoms and linear rotors. *J. Chem. Phys.* **108**: 5265–5280.
39. Maergoiz AI, Nikitin EE, Troe J, Ushakov VG. (1998) Classical trajectory and statistical adiabatic channel study of the dynamics of capture and unimolecular bond fission. V. Valence interactions between two linear rotors. *J. Chem. Phys.* **108**: 9987–9998.
40. Quack M, Troe J. (1975) Complex formation in reactive and inelastic scattering. Statistical adiabatic channel model of unimolecular processes. III. *Ber. Bunsenges. Phys. Chem.* **79**: 170–183.
41. Garrett BC, Truhlar DG. (1979) Generalized transition state theory. Quantum effects for collinear reactions of hydrogen molecules and isotopically substituted hydrogen molecules. *J. Phys. Chem.* **83**: 1052–1079.
42. Clary DC. (1984) Rates of chemical reactions dominated by long-range intermolecular forces. *Mol. Phys.* **53**: 3–21.
43. Stoecklin T, Dateo CE, Clary DC. (1991) Rate constant calculations on fast diatom diatom reactions. *J. Chem. Soc. Faraday Trans.* **87**: 1667–1679.
44. Stoecklin T, Clary DC. (1992) Fast reactions between diatomic and polyatomic molecules. *J. Phys. Chem.* **96**: 7346–7351.

45. Stoecklin T, Clary DC. (1995) Fast reactions between a linear molecule and a polar symmetrical top. *Theochem. J. Mol. Struc.* **341**: 53–61.
46. Clary DC, Henshaw JP. (1987) Chemical reactions dominated by long-range intermolecular forces. *Faraday Disc.* **84**: 333–349.
47. Clary DC. (1987) Rate constants for the reactions of ions with dipolar polyatomic molecules. *J. Chem. Soc. Faraday Trans. II*, **83**: 139–148.
48. Clary DC, Smith D, Adams NG. (1985) Temperature dependence of rate coefficients for reactions of ions with dipolar molecules. *Chem. Phys. Lett.* **119**: 320–326.
49. Clary DC, Dateo CE, Smith D. (1990) Rates for the reactions of open-shell ions with molecules. *Chem. Phys. Lett.* **167**: 1–6.
50. Gonzalez AI, Clary DC, Yanez, M. (1997) Calculations of rate constants for reactions of first and second row cations. *Theor. Chem. Acc.* **98**: 33–41.
51. Dateo CE, Clary DC. (1989) Rate constant calculations on the $C^+ + HCl$ reaction. *J. Chem. Phys.* **90**: 7216–7228.
52. Graff MM, Wagner AF. (1990) Theoretical studies of fine-structure effects and long-range forces: Potential-energy surfaces and reactivity of $O(^3P) + OH(^2\Pi)$. *J. Chem. Phys.* **92**: 2423–2439.
53. Clary DC, Stoecklin TS, Wickham AG. (1993) Rate constants for chemical-reactions of radicals at low-temperatures. *J. Chem. Soc. Faraday Trans.* **89**: 2185–2191.
54. Reignier D, Stoecklin T. (1999) Comparison of the spin-orbit selectivity of reaction involving atoms in a 2P or 3P state and a linear molecule in a Π or Σ state at very low temperature. *Chem. Phys. Lett.* **303**: 576–582.
55. Clary DC, Werner HJ. (1984) Quantum calculations on the rate constant for the $O + OH$ reaction. *Chem. Phys. Lett.* **112**: 346–350.
56. Troe J. (1987) Statistical adiabatic channel model for ion-molecule capture processes. *J. Chem. Phys.* **87**: 2773–2780.
57. Maergoiz AI, Nikitin EE, Troe J. (1991) Adiabatic channel potential curves for two linear dipole rotors. I. Classification of states and numerical calculations for identical rotors. *J. Chem. Phys.* **95**: 5117–5127.
58. Ramillon M, McCarroll R. (1994) Adiabatic capture models for fast chemical reactions. *J. Chem. Phys.* **101**: 8697–8699.
59. Sakimoto K. (1982) Perturbed rotational state method for collisions between an ion and an asymmetric-top rigid rotor. *Chem. Phys.* **68**: 155–170.
60. Bates DR, Morgan WL. (1987) Adiabatic invariance treatment of hitting collisions between ions and symmetrical top dipolar molecules. *J. Chem. Phys.* **87**: 2611–2616.
61. Maergoiz AI, Nikitin EE, Troe J. (1995) Statistical adiabatic channel calculation of accurate low-temperature rate constants for the recombination of OH radicals in their ground rovibronic state. *J. Chem. Phys.* **103**: 2083–2091.
62. Vogt E, Wannier GH. (1954) Scattering of ions by polarization forces. *Phys. Rev.* **95**: 1190–1198.

63. Hessler JP. (1999) New empirical rate expression for reactions without a barrier: Analysis of the reaction of CN with O₂. *J. Chem. Phys.* **111**: 4068–4076.
64. Dashevskaya EI, Maergoiz AI, Troe J, Litvin I, Nikitin EE. (2003) Low-temperature behavior of capture rate constants for inverse power potentials. *J. Chem. Phys.* **118**: 7313–7320.
65. Nikitin EE, Troe J. (2005) Dynamics of ion-molecule complex formation at very low energies and temperatures. *Phys. Chem. Chem. Phys.* **7**: 1540–1551.
66. Nikitin EE, Troe J, Ushakov VG. (1995) Adiabatic and postadiabatic channel description of atom-diatom long-range half-collision dynamics: Interchannel radial coupling for P₁ and P₂ anisotropy. *J. Chem. Phys.* **102**: 4101–4111.
67. Klippenstein SJ. (1992) Variational optimizations in RRKM theory calculations for unimolecular dissociations with no reverse barrier. *J. Chem. Phys.* **96**: 367–371.
68. Olkhov RV, Smith IWM. (2007) Rate coefficients for reaction and for rotational energy transfer in collisions between CN in selected rotational levels ($X^2\Sigma^+$, $v = 2$, $N = 0, 1, 6, 10, 15$, and 20) and C₂H₂. *J. Chem. Phys.* **126**: 134314; 1–10.
69. Troe J. (1996) Statistical adiabatic channel model for ion-molecule capture processes. II. Analytical treatment of ion-dipole capture. *J. Chem. Phys.* **105**: 6249–6262.
70. Smith D, Adams NG, Miller TM. (1978) Laboratory study of reactions of N⁺, N₂⁺, N₃⁺, N₄⁺, O⁺, O₂⁺, and NO⁺ ions with several molecules at 300 K. *J. Chem. Phys.* **69**: 308–318.
71. Marquette JB, Rowe BR, Dupeyrat G, Poissant G, Rebrion C. (1985) Ion-polar-molecule reactions: A CRESU study of He⁺, C⁺, N⁺, + H₂O, NH₃ at 27, 68, and 163 K. *Chem. Phys. Lett.* **122**: 431–435.
72. Smith IWM, Sage AM, Donahue NM, Herbst E, Park IH. (2006) The temperature-dependence of rapid low temperature reactions: experiment, understanding and prediction. *Faraday Disc.* **133**: 137–156.
73. Canosa A, Sims IR, Travers D, Smith IWM, Rowe BR. (1997). Reactions of the methyldene radical with CH₄, C₂H₂, C₂H₆, and but-1-ene studied between 23 and 295 K with a CRESU apparatus. *Astron. Astrophys.* **323**: 644–651.
74. Berman MR, Fleming JW, Harvey AB, Lin MC. (1982) Temperature dependence of the reactions of CH radicals with unsaturated hydrocarbons. *Chem. Phys.* **73**: 27–33.
75. Butler JE, Fleming JW, Goss LP, Lin MC. (1981) Kinetics of CH radical reactions with selected molecules at room temperature. *Chem. Phys.* **56**: 355–365.
76. Tan X, Dong F, Li X. (1998) Importance of different multipole interactions in fast reactions at low temperature. *J. Phys. Chem. A.* **102**: 8169–8175.
77. Le Picard SD, Canosa A, Geppert W, Stoecklin T. (2004) Experimental and theoretical temperature dependence of the rate coefficient of the

- $B(^2P_{1/2,3/2}) + O_2(X^3\Sigma_g^-)$ reaction in the [24–295 K] temperature range. *Chem. Phys. Lett.* **385**: 502–506.
78. Le Picard SD, Canosa A, Travers D, Chastaing D, Rowe BR, Stoecklin T. (1997) Experimental and theoretical kinetics for the reaction of Al with O_2 at temperatures between 23 and 295 K. *J. Phys. Chem. A*. **101**: 9988–9992.
 79. Le Picard SD, Canosa A, Reignier D, Stoecklin T. (2002) A comparative study of the reactivity of the silicon atoms $Si(^3P_J)$ towards O_2 and NO molecules at very low temperature. *Phys. Chem. Chem. Phys.* **4**: 3659–3664.
 80. Andersson S, Markovic N, Nyman G. (2003) Computational studies of the kinetics of $C + NO$ and $O + CN$ reactions. *J. Phys. Chem. A*. **107**: 5439–5447.
 81. Chastaing D, Le Picard SD, Sims IR. (2000) Direct kinetic measurements on reactions of atomic carbon, $C(^3P)$, with O_2 and NO at temperatures down to 15 K. *J. Chem. Phys.* **112**: 8466–8469.
 82. Beghin A, Stoecklin T, Rayez JC. (1995) Rate constant calculations for atom-diatom reactions involving an open-shell atom and a molecule in a Π electronic state: Application to the $C(^3P) + NO(X^2\Pi)$ reaction. *Chem. Phys.* **195**: 259–270.
 83. Geppert WD, Goulay F, Naulin C, Costes M, Canosa A, Le Picard SD, Rowe BR. (2004) Rate coefficients and integral cross-sections for the reaction of $B(^2P_J)$ atoms with acetylene. *Phys. Chem. Chem. Phys.* **6**: 566–571.
 84. Canosa A, Le Picard SD, Geppert WD. (2004) Experimental kinetics study of the reaction of boron atoms, $B(^2P_J)$, with ethylene at very low temperatures (23–295 K). *J. Phys. Chem. A*. **108**: 6183–6185.
 85. Chastaing D, James PL, Sims IR, Smith IWM. (1999) Neutral-neutral reactions at the temperatures of interstellar clouds: Rate coefficients for reactions of atomic carbon, $C(^3P)$, with O_2 , C_2H_2 , C_2H_4 and C_3H_6 down to 15 K. *Phys. Chem. Chem. Phys.* **1**: 2247–2256.
 86. Chastaing D, Le Picard SD, Sims IR, Smith IWM. (2001) Rate coefficients for the reactions of $C(^3P_J)$ atoms with C_2H_2 , C_2H_4 , $CH_3C\equiv CH$ and $H_2C=C=CH_2$ at temperatures down to 15 K. *Astron. Astrophys.* **365**: 241–247.
 87. Canosa A, Le Picard SD, Gougeon S, Rebrion-Rowe C, Travers D, Rowe BR. (2001) Rate coefficients for the reactions of $Si(^3P_J)$ with C_2H_2 and C_2H_4 : Experimental results down to 15 K. *J. Chem. Phys.* **115**: 6495–6503.
 88. Duagey N, Caubet P, Retail B, Costes M, Bergeat A, Dorthe G. (2005) Kinetic measurements on methylidyne radical reactions with several hydrocarbons at low temperatures. *Phys. Chem. Chem. Phys.* **7**: 2921–2927.
 89. Goulay F, Rebrion-Rowe C, Biennier L, Le Picard SD, Canosa A, Rowe BR. (2006) Reaction of anthracene with CH radicals: An experimental study of the kinetics between 58 and 470 K. *J. Phys. Chem. A*. **110**: 3132–3137.
 90. Sims IR, Queffelec JL, Defrance A, Rebrion-Rowe C, Travers D, Rowe BR, Herbert LB, Karthausier J, Smith IWM. (1993) Rate constants for the reactions of CN with hydrocarbons at low and ultra-low temperatures. *Chem. Phys. Lett.* **211**: 461–468.

91. Carty D, Le Page V, Sims IR, Smith IWM. (2001) Low temperature rate coefficients for the reactions of CN and C₂H radicals with allene (CH₂=C=CH₂) and methyl acetylene (CH₃C ≡ CH). *Chem. Phys. Lett.* **344**: 310–316.
92. Opansky B, Leone SR. (1996) Rate coefficients of C₂H with C₂H₄, C₂H₆, and H₂ from 150 to 359 K. *J. Phys. Chem.* **100**: 19904–19910.
93. Chastaing D, James PL, Sims IR, Smith IWM. (1998) Neutral-neutral reactions at the temperatures of interstellar clouds. Rate coefficients for reactions of C₂H radicals with O₂, C₂H₂, C₂H₄, and C₃H₆ down to 15 K. *Faraday Disc.* **109**: 165–181.
94. Hoobler RJ, Leone SR. (1999) Low-temperature rate coefficients for reactions of the ethynyl radical (C₂H) with C₃H₄ isomers methylacetylene and allene. *J. Phys. Chem. A.* **103**: 1342–1346.
95. Vakhtin AB, Heard DE, Smith IWM, Leone SR. (2001) Kinetics of C₂H radical reactions with ethene propene and 1-butene measured in a pulsed Laval nozzle apparatus at T = 103 and 296 K. *Chem. Phys. Lett.* **348**: 21–26.
96. Harding LB, Klippenstein SJ, Jasper AW. (2007) *Ab initio* methods for reactive potential surfaces. *Phys. Chem. Chem. Phys.* **9**: 4055–4070.
97. Andersson K, Malqvist PA, Roos BO. (1992) 2nd-order perturbation theory with a complete active space self consistent field reference function. *J. Chem. Phys.* **96**: 1218–1226.
98. MOLPRO is a package of programs written by Werner HJ, Knowles PJ, with contributions from Almlof J, Amos RD, Berning A, Cooper DL, Deegan MJO, Dobbyn AJ, Eckert F, Elbert ST, Hampel C, Lindh R, Lloyd AW, Meyer W, Nicklass A, Peterson K, Pitzer R, Stone AJ, Taylor PR, Mura ME, Pulay P, Schutz M, Stoll H, Thorsteinsson T.
99. Kendall RA, Dunning TH, Harrison RJ. (1992) Electron affinities of the 1st-row atoms revisited. Systematic basis sets and correlated wave functions. *J. Chem. Phys.* **96**: 6796–6806.
100. Simandiras ED, Rice JE, Lee TJ, Amos RD, Handy NC. (1988) On the necessity of f basis functions for bending frequencies. *J. Chem. Phys.* **88**: 3187–3195.
101. Raghavachari K, Trucks GW, Pople JA, Head-Gordon M. (1989) A 5th order perturbation comparison of electronic structure theories. *Chem. Phys. Lett.* **157**: 479–483.
102. Pople JA, Head-Gordon M, Raghavachari K. (1987) Quadratic configuration interaction. A general technique for determining electron correlation energies. *J. Chem. Phys.* **87**: 5968–5975.
103. Lee TJ, Taylor PR. (1989) A diagnostic for determining the quality of single reference electron correlation methods. *Int. J. Quantum. Chem.* **S23**: 199–207.
104. Curtiss LA, Raghavachari K, Redfern PC, Rassolov V, Pople JA. (1998) Gaussian-3 (G3) theory for molecules containing first and second row atoms. *J. Chem. Phys.* **109**: 7764–7776.
105. Martin JML, Uzan O. (1998) Basis set convergence in second-row compounds. The importance of core polarization functions. *Chem. Phys. Lett.* **282**: 16–24.

106. Tajti A, Szalay PG, Csaszar AG, Kallay M, Gauss J, Valeev EF, Flowers BA, Vazquez J, Stanton JF. (2004) High accuracy extrapolated *ab initio* thermochemistry. *J. Chem. Phys.* **121**: 11599–11613.
107. Karton A, Rabinovich E, Martin JML, Ruscic B. (2006) W4 theory for computational thermochemistry: In pursuit of confident sub-kJ/mol predictions. *J. Chem. Phys.* **125**: 144108: 1–17.
108. Curtiss LA, Redfern PC, Raghavachari K. (2007) Gaussian-4 theory. *J. Chem. Phys.* **126**: 084108: 1–12.
109. Grimme S, Antony J, Schwabe T, Muck-Lichtenfeld C. (2007) Density functional theory with dispersion corrections for supramolecular structures, aggregates, and complexes of bio(organic) molecules. *Org. Biomol. Chem.* **5**: 741–758.
110. Harding LB, Georgievskii Y, Klippenstein SJ. (2005) Predictive theory for hydrogen atom-hydrocarbon radical association kinetics. *J. Phys. Chem. A.* **109**: 4646–4656.
111. Klippenstein SJ, Georgievskii Y, Harding LB. (2006) Predictive theory for the association kinetics of two alkyl radicals. *Phys. Chem. Chem. Phys.* **8**: 1133–1147.
112. Harding LB, Klippenstein SJ, Georgievskii Y. (2007) On the combination reactions of hydrogen atoms with resonance stabilized hydrocarbon radicals. *J. Phys. Chem. A.* **111**: 3789–3801.
113. Jasper AW, Klippenstein SJ, Harding LB, Ruscic B. (2007) Kinetics of the reaction of methyl radical with hydroxyl radical and methanol decomposition. *J. Phys. Chem. A.* **111**: 3932–3950.
114. Herbst E, DeFrees DJ, Talbi D, Pauzat F, Koch W, McLean AD. (1991) Calculations on the rate of the ion-molecule reaction between NH_3^+ and H_2 . *J. Chem. Phys.* **94**: 7842–7848.
115. Herbst E. (1994) Tunneling in the $\text{C}_2\text{H}-\text{H}_2$ reaction at low temperature. *Chem. Phys. Lett.* **222**: 297–301.
116. Miller WH, (1976) Unified statistical model for “complex” and “direct” reaction mechanisms. *J. Chem. Phys.* **65**: 2216–2223.
117. Miller JA, Klippenstein SJ. (2006) Master equation methods in gas phase chemical kinetics. *J. Phys. Chem. A.* **110**: 10528–10544.
118. Singleton DL, Cvetanovic RJ. (1976) Temperature dependence of the reactions of oxygen atoms with olefins. *J. Am. Chem. Soc.* **98**: 6812–6819.
119. Clary DC, Nyman G, Hernandez R. (1994) Mode selective chemistry in the reactions of OH with HBr and HCl. *J. Chem. Phys.* **101**: 3704–3714.
120. Faure A, Rist C, Valiron P. (1999) Temperature dependence for the $\text{CN} + \text{NH}_3$ reaction under interstellar conditions: Beyond capture theories? *Astron. Astrophys.* **348**: 972–977.
121. Faure A, Wiesenfeld L, Valiron P. (2000) Temperature dependence of fast neutral-neutral reactions: A triatomic model study. *Chem. Phys.* **254**: 49–67.
122. Herbst E, Woon DE. (1997) The rate of the reaction between C_2H and C_2H_2 at interstellar temperatures. *Astrophys. J.* **489**: 109–112.

123. Clary DC, Haider N, Husain D, Kabir M. (1994) Interstellar carbon chemistry: Reaction rates of neutral atomic carbon with organic molecules. *Astrophys. J.* **422**: 416–422.
124. Buonomo E, Clary DC. (2001). A quantum study on the reaction between $C(^3P)$ and acetylene. *J. Phys. Chem. A.* **105**: 2694–2707.
125. Chen Y, Rauk A, Tschuikow-Roux E. (1991) On the question of negative activation energies: Absolute rate constants by RRKM and G1 theory for $CH_3 + HX \rightarrow CH_4 + X$ ($X = Cl, Br$) reactions. *J. Phys. Chem.* **95**: 9900–9908.
126. Jodkowski JT, Rayez MT, Rayez JC, Berces T, Dobe S. (1998) Theoretical study of the kinetics of the hydrogen abstraction from methanol. 3. Reaction of methanol hydrogen atom, methyl, and hydroxyl radicals. *J. Phys. Chem. A.* **103**: 3750–3765.
127. Xu SC, Zhu RS, Lin MC. (2006) *Ab initio* study of the $OH + CH_2O$ reaction: The effect of the $OH-OCH_2$ complex on the H-abstraction kinetics. *Int. J. Chem. Kinet.* **38**: 322–326.
128. Krasnoperov LV, Peng J, Marshall P. (2006) Modified transition state theory and negative apparent activation energies of simple metathesis reactions: Application to the reaction $CH_3 + HBr \rightarrow CH_4 + Br$. *J. Phys. Chem. A.* **110**: 3110–3120.
129. Yu J, Klippenstein SJ. (1991) Variational calculation of the rate of dissociation of CH_2CO in 1CH_2 and CO on an *ab initio* determined potential energy surface. *J. Phys. Chem.* **95**: 9882–9889.
130. Klippenstein SJ, Marcus RA. (1989) Application of unimolecular reaction rate theory for highly flexible transition states to the dissociation of CH_2CO into CH_2 and CO. *J. Chem. Phys.* **91**: 2280–2292.
131. Klippenstein SJ, East ALL, Allen WD. (1996) A high level *ab initio* map and direct statistical treatment of the fragmentation of singlet ketene. *J. Chem. Phys.* **105**: 118–140.
132. Matsumoto K, Klippenstein SJ, Tonokura K, Koshi M. (2005) Channel specific rate constants relevant to the thermal decomposition of disilane. *J. Phys. Chem. A.* **109**: 4911–4920.
133. Greenwald EE, North SW, Georgievskii Y, Klippenstein SJ. (2007) A two transition state model for radical-molecule reactions: Applications to isomeric branching in the OH-isoprene reaction. *J. Phys. Chem. A.* **111**: 5582–5592.
134. Bullock GE, Cooper RE. (1972) Reactions of cyanogen radicals. Part 3. Arrhenius parameters for reactions with alkanes. *J. Chem. Soc. Faraday Trans. I* **68**: 2185–2190.
135. Hess WP, Durant JL, Tully FP. (1989) Kinetic study of the reactions of CN with ethane and propane. *J. Phys. Chem.* **93**: 6402–6407.
136. Balla RJ, Castleton KH. (1991) Absolute rate constants for the reaction of CN with CH_4 , C_2H_6 , and C_3H_8 from 292 to 1500 K using high temperature photochemistry and diode laser absorption. *J. Phys. Chem.* **95**: 8694–8701.
137. Atakan B, Wolfrum J. (1991) Kinetic studies of the reactions of CN radicals with alkanes in the temperature range between 294 and 1260 K. *Chem. Phys. Lett.* **186**: 547–552.

138. Herbert L, Smith IWM, Spencer-Smith RD. (1992) Rate constants for the elementary reactions between CN radicals and CH₄, C₂H₆, C₂H₄, C₃H₆, and C₂H₂ in the range 295 to 700 K. *Int. J. Chem. Kinet.* **24**: 791–802.
139. Zellner R, Lorenz K. (1984) Laser photolysis/resonance fluorescence study of the rate constants for the reactions of OH radicals with C₂H₄ and C₃H₆. *J. Phys. Chem.* **88**: 984–989.
140. Klein T, Barnes I, Becker KH, Fink EH, Zabel F. (1984) Pressure dependence of the rate constants for the reactions of C₂H₄ and C₃H₆ with OH radicals at 295 K. *J. Phys. Chem.* **88**: 5020–5025.
141. Kuo CH, Lee YP. (1991) Kinetics of the reaction OH + C₂H₄ in He, N₂, and O₂ at low pressure. *J. Phys. Chem.* **95**: 1253–1257.
142. Vakhtin AB, Murphy JE, Leone SR. (2003) Low temperature kinetics of reactions of OH radical with ethene, propene, and 1-butene. *J. Phys. Chem. A*. **107**: 10055–10062.
143. Cleary PA, BaezaRomero MT, Blitz MA, Heard DE, Pilling MJ, Seakins PW, Wang L. (2006) Determination of the temperature and pressure dependence of the reaction OH + C₂H₄ from 200–400 K using experimental and master equation analyses. *Phys. Chem. Chem. Phys.* **8**: 5633–5642.
144. Taylor SE, Goddard A, Blitz MA, Cleary PA, Heard DE. (2008) Pulsed Laval nozzle study of the kinetics of OH with unsaturated hydrocarbons at very low temperatures. *Phys. Chem. Chem. Phys.* **10**: 422–429.
145. Chesnavich WJ, Bass L, Su T, Bowers MT. (1981) Multiple transition states in unimolecular reactions: A transition state switching model. Application to the C₄H₈⁺ system. *J. Chem. Phys.* **74**: 2228–2246.
146. Jarrold MF, Bass LM, Kemper PR, van Koppen PAM, Bowers MT. (1983) Unimolecular and bimolecular reactions in the C₄H₆⁺ system. Experiment and theory. *J. Chem. Phys.* **78**: 3756–3766.
147. Jarrold MF, Wagner-Redeker W, Illies AJ, Kirchner NJ, Bowers MT. (1984) Unimolecular and bimolecular reactions in the C₆H₆⁺ system. Experiment and theory. *Int. J. Mass Spec. Ion Proc.* **58**: 63–95.
148. Georgievskii Y, Klippenstein SJ. (2003) Transition state theory for multi-channel addition reactions: Multifaceted dividing surfaces. *J. Phys. Chem. A*. **107**: 9776–9781.
149. Klippenstein SJ, Khundkar LR, Zewail AH, Marcus RA. (1988) Application of unimolecular reaction rate theory for highly flexible transition states to the dissociation of NCNO into NC and NO. *J. Chem. Phys.* **89**: 4761–4770.
150. Reznickova JG, Hippler H, Striebel F, Tevazdze L. (2000) A saturated LIF study on the high pressure limiting rate constant of the reaction CN + NO + M → NCNO + M between 200 and 600 K. *Z. Phys. Chem.* **214**: 1115–1136.
151. Smith GP. (2003) Rate theory of methyl recombination at the low temperatures and pressures of planetary atmospheres. *Chem. Phys. Lett.* **376**: 381–388.
152. Sims IR, Smith IWM. (1988) Rate constants for the radical-radical reaction between CN and O₂ at temperatures down to 99 K. *Chem. Phys. Lett.* **151**: 481–484.

153. Sims IR, Queffelec JL, Defrance A, Rebrion-Rowe C, Travers D, Rowe BR, Smith IWM. (1992) Ultra-low temperature kinetics of neutral-neutral reactions: The reaction of $\text{CN} + \text{O}_2$ down to 26 K. *J. Chem. Phys.* **97**: 8798–8800.
154. Sims IR, Queffelec JL, Defrance A, Rebrion-Rowe C, Travers D, Bocherel P, Rowe BR, Smith IWM. (1994) Ultralow temperature kinetics of neutral-neutral reactions. The technique and results for the reactions $\text{CN} + \text{O}_2$ down to 13 K and $\text{CN} + \text{NH}_3$ down to 25 K. *J. Chem. Phys.* **100**: 4229–4241.
155. Edvardsson D, Williams CF, Clary DC. (2006) Rate constant calculations on the $\text{N}(^4\text{S}) + \text{OH}(^2\Pi)$ reaction. *Chem. Phys. Lett.* **431**: 261–266.
156. Xu C, Xie D, Honvault P, Lin SY, Guo H. (2007) Rate constant for $\text{OH}(^2\Pi) + \text{O}(^3\text{P}) \rightarrow \text{H}(^2\text{S}) + \text{O}_2(^3\Sigma_g^-)$ reaction on an improved ab initio potential energy surface and implications for the interstellar oxygen problem. *J. Chem. Phys.* **127**: 024304; 1–6.
157. Dayou F, Spielfiedel A. (2003) *Ab initio* calculation of the ground ($1\text{A}'$) potential energy surface and theoretical rate constant for the $\text{Si} + \text{O}_2 \rightarrow \text{SiO} + \text{O}$ reaction. *J. Chem. Phys.* **119**: 4237–4250.
158. Zanchet A, Halvick P, Rayez JC, Bussery-Honvault B, Honvault P. (2007) Cross sections and rate constants for the $\text{C}(^3\text{P}) + \text{OH}(^2\Pi) \rightarrow \text{CO}(^1\Sigma^+) + \text{H}(^2\text{S})$ reaction using a quasiclassical trajectory method. *J. Chem. Phys.* **126**: 184308; 1–6.
159. Tashiro M, Schinke R. (2003) The effect of spin-orbit coupling in complex forming $\text{O}(^3\text{P}) + \text{O}_2$ collisions. *J. Chem. Phys.* **119**: 10186–10193.
160. Yagi K, Takayanagi T, Taketsugu T, Hirao K. (2004) The effect of spin-orbit coupling on fast neutral chemical reaction $\text{O}(^3\text{P}) + \text{CH}_3 \rightarrow \text{CH}_3\text{O}$. *J. Chem. Phys.* **120**: 10395–10403.